

Department of Mechanical and Mechatronics Engineering

**UNIVERSITY OF
WATERLOO**



ME 482 - Mechanical Engineering Design Project

Final Design Report: PackerPro

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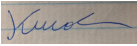
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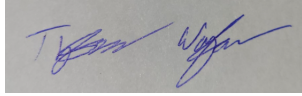
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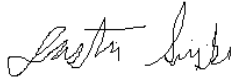
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1.0 Summary of Work Completed in ME481

1.1 Background and Motivation

Richters Herbs is a small Canadian company that sells herbs, seeds and other botanical related products to North Americans. To reduce cost, Richters had expressed a desire to automate certain tasks which currently require manual labour. Specifically, the aim of this project was to develop a solution which automates the seed packing process.

1.1.1 Need Statement

A need exists to process bulk seeds into packages of smaller quantities, with minimal manual interaction.

1.1.2 Project Requirements

Following the need statement, the group had outlined several key requirements which must be met. Beginning with fill accuracy, the robot must be able to fill each packet with the appropriate amount of seeds (based on weight). Given a target weight, the robot must fill to within 10 wt. % . To benefit Richters, the robot also must produce packets as fast or faster than the current manual rate of 15 packets per minute. Additionally, the set-up time must be minimized to ensure the total automated process is faster than manual.

The robot must also be able to vary its production output depending on the current needs. Anywhere between 20 and 100 packets must be able to be produced in a given run. This requirement corresponds with the required hopper storage. The hopper must be able to store approximately 1.5 L of seeds to be able to fill the required amount of packets without refilling. While packing, it is also important to minimize possible damage to the seeds. Seeds must never exceed around 25°C in internal temperature.

During the production runs the robot must maintain a strict failure rate. The robot must package a minimum of 999 out of every 1000 packets properly. After a run is complete the robot's internals must be cleared of any debris or contaminants. This feature should be automated or be a

simple manual interaction. Finally, the robot must be easy to maintain while maintaining its functionality. This includes a level of corrosion and general wear resistance built into the final design.

1.2 Proposed Solution

This engineering problem was to be solved with a mechanism that is capable of taking an input of bulk seeds and empty seed packets and outputting full, sealed packets. The system was to be comprised of 5 different subsystems, which will be described in greater detail in section 2.0 of this report. These subsystems would be the packet injection, rotating base, seed portioner, heat sealer and packet ejection [1]. As a whole, the subsystems will work together to deposit and seal a measured portion of the bulk seed inside an empty packet, with no contamination of foreign seeds.

1.3 Initial Design

At the end of ME481 the design on CAD was conceptually similar to the final design however many small changes were made to the subsystems. The previous design also had a rotary table which transported the packets to the different subsystems. The four subsystems and their functions also remained the same, packet injection, seed portioning, heat sealing the packet, and packet ejection.

The packet injection would be powered by a pneumatic cylinder and the individual packets would be attached with suction cups. Once the packet was inserted in the machine it would be held by a packet clamp with a spring loaded arm to hold the packet.

Seed portioner originally contained a custom seed hopper that fed into a rotating auger. The auger would be rotated by a stepper motor at one end and would drop the seeds through a spout at the other end. At this point in the design a mechanism to spread the packets before filling was not yet included.

The original Packet sealer design was functionally identical to the final design. It contained a pneumatic cylinder that would pinch the top of the packet between a hot resistive element heater and a piece of steel to melt the adhesive and seal the packet.

The original packet ejection mechanism was a simple pneumatic cylinder that would push the packet clamp open and allow the finished packet to fall out of the machine.

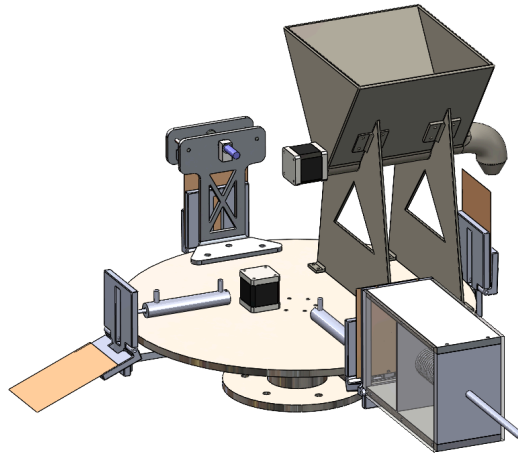


Figure 1: Initial system design.

1.4 Verification of Concepts

In ME481, three critical subsystems were prototyped to ensure sufficient conceptual functionality. The first of these critical subsystems is the seed portioner. A vibratory and auger type portioner were tested, with the auger system proving a better fit. The auger provides consistent and accurate denominations of seeds which can be controlled easily through a direct-drive mounted stepper motor. The next critical subsystem that was tested was that of the packet feeder. The ability of a pneumatic suction cup to pull a packet through a slot was verified by the tests. It was shown to be a reliable and simple system. The last critical subsystem tested was the packet opener subsystem. A pneumatic suction cup design was tested. This design met all requirements, opening packets repeatedly without failure. The simple and reliable suction cups were easily controlled by applying negative pressure with a vacuum pump.

2.0 Design Work Completed

2.1 Standard Design Practices

Since each member of the group participated in the design of the seed packaging system, some basic design standards were set to ensure a uniform overall design once the subsystems were manufactured and assembled together. Standard material choices were limited to 6061 aluminum and black PLA plastic. Aluminum was desired to keep the overall weight of the machine down while maintaining good corrosion resistance and great structural rigidity.

Wherever possible, components were to be designed to be manufactured by the waterjet in the E5 student engineering shop. Therefore, each part had to fit within a 12 inch square area and have a maximum thickness of .375 inches. Using the student waterjet reduced costs as well as the machining time required to make pieces, additionally, it resulted in the ability to produce parts to a higher degree of finish with the inclusion of the team logo. However, if the parts required 3d features or could not conform to these size restraints, other manufacturing methods were permitted. For small, non-structural components of complex, 3d geometry, the design was to be based on the use of a 3d printer for manufacturing using black PLA filament. This allowed for quick and inexpensive prototyping and final manufacturing of components beyond the ability of our skills or tools to machine.

The design work was to be completed in metric units with the standard fastener of choice being black oxide, socket head cap screws. This type of screw was chosen for its small head clearance requirements. The black oxide finish was selected to keep costs down, its mild corrosion resistance was acceptable for the dry working environment the machine was to be operated in.

2.2 Final System Design Overview

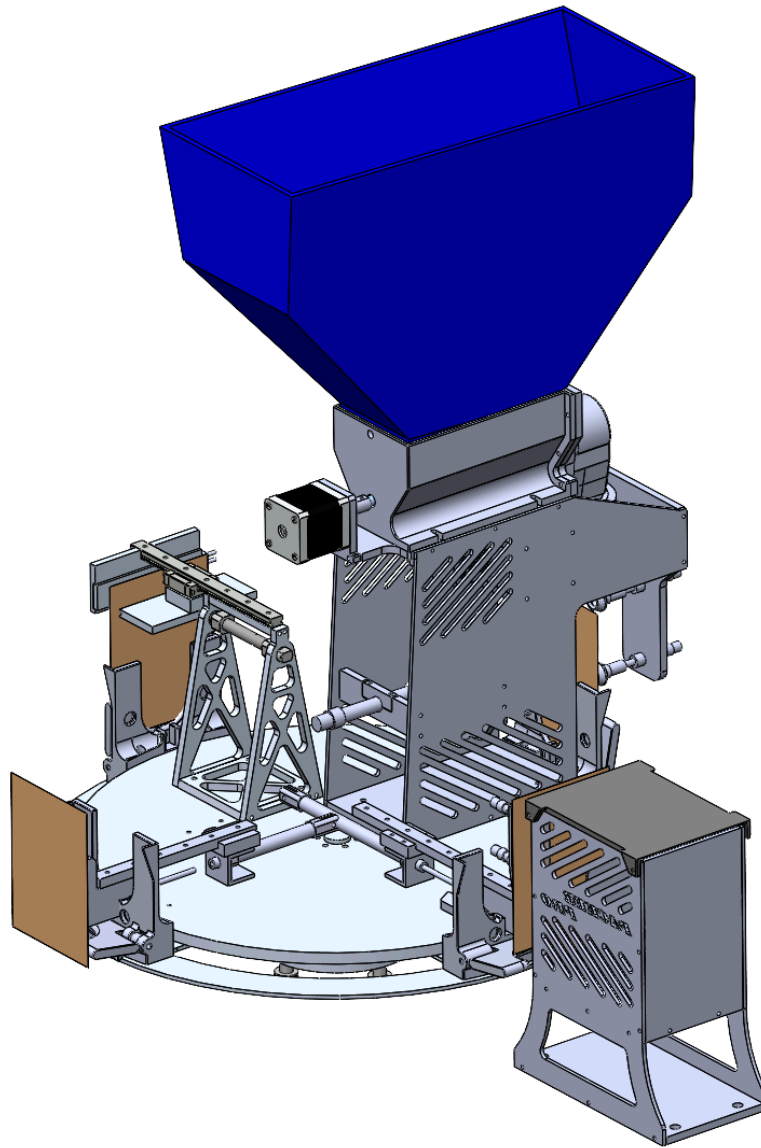


Figure 2: Final system design.

Our design consists of a rotary table that rotates four packets held in individual clamps around a stationary plate that mounts the four subsystems mounted 90 degrees from each other. The subsystems include “Packet Injection” which pulls single packets from a stack into the machine, “Seed Portioning Tower” which spreads the packets and drops a portioned amount of seeds into it, “Heat Sealer” which melts an airtight seal into the top of the packet, and “Packet Ejection” which ejects the packets from the system. The motion is controlled with pneumatic cylinders, servos, and stepper motors.

2.3 Rotating Base

The rotating base subsystem consisted of 3 main components: the packet clamp, the stationary structure and the rotating plate. These three components work together to both support the entire system and transport the empty seed packets between subsystems.

2.3.1 Packet Clamp

The “Packet Clamp” is what interfaces between the seed packets and our machine, it holds the packets in place as it rotates around the base and allows the packets to come in and out of the machine. The packet clamp bolts into the stationary plate and there are 4 of them total spaced 90 degrees apart.

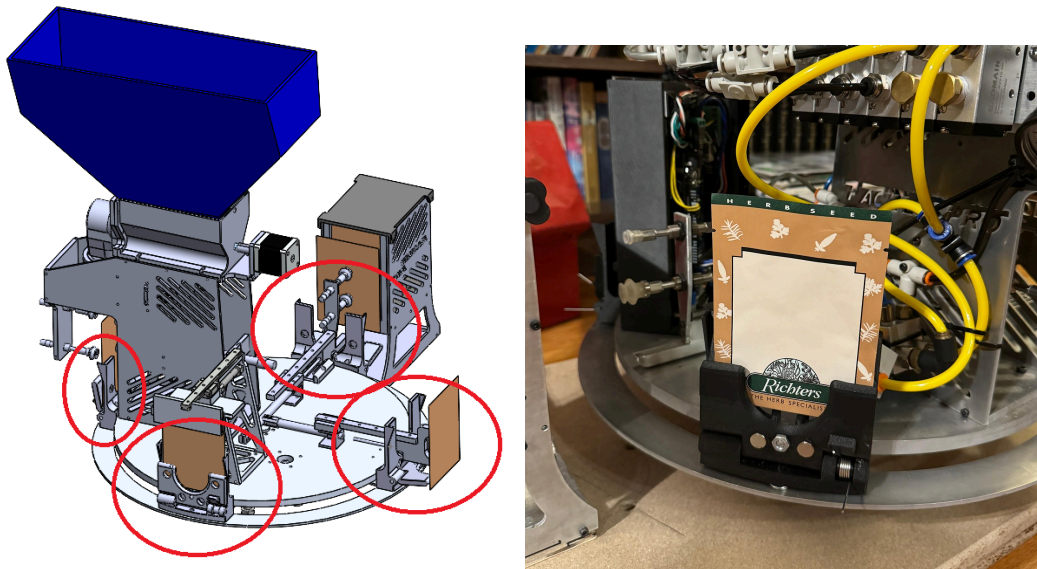


Figure 3: Packet clamp layout on rotating plate.

2.3.2 Packet Clamp Initial Design

The initial “packet Clamp” design consisted of two 3D printed plastic parts, the stationary “Back Plate” that bolts into the rotating plate and the clamp which rotates about a steel axle to pinch the packet in place. Inside the hinge on both sides there is a torsion spring forcing the clamp closed at all times. The problem with this design is the clamping force when the clamp is in the closed position is very small because the spring is close to its resting angle of 90 degrees. To fix this we pretensioned the spring by decreasing the angle of the spring when the clamp is in the upright

position. However when the clamp was in the fully open position (during packet injection) the clamping force was so great that it would bend the pneumatic cylinder piston causing it to seize. This presented an important design criteria for the packet clamp which is to have a minimum clamping force when in fully open position, and max clamping force when in upright position.

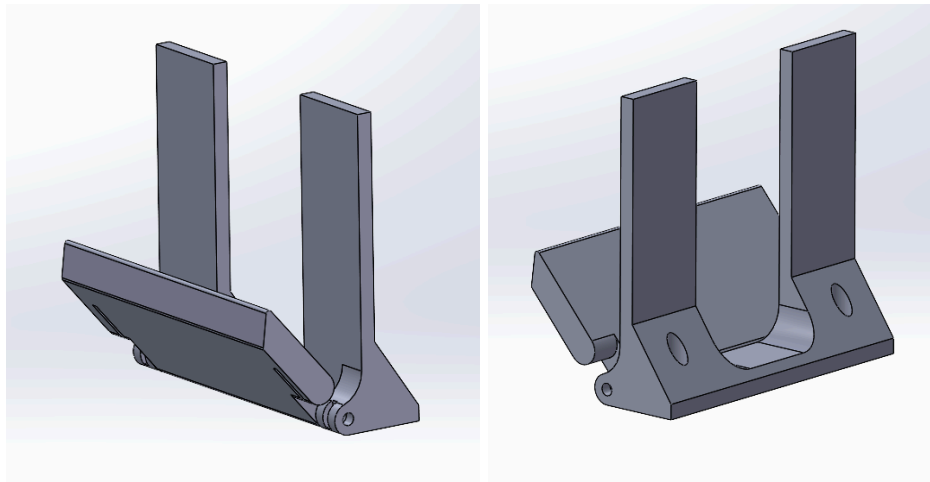


Figure 4: Initial packet clamp design.

2.3.3 Packet Clamp Final Design

The final packet clamp design in order to maximize the clamping force in closed position without increasing the clamping force in the open position added 4 magnets clamping the two pieces closed. The nature of magnets makes this magnetic clamping force at a maximum when the magnet pairs are closest together and almost negligible when they are far apart in the open position. The 8 cylindrical magnets are pressed into the plastic and then epoxied in place. The magnets generated enough force when closed that we could eliminate one of the torsion springs resulting in a smaller closing force especially at the fully open position. A 3D printed cylindrical spacer was added to help the rotation of the torsional spring. The ID of the torsional spring is much greater than the OD of the steel axil in the initial design; this would cause the torsional spring to bind up.

The back plate has a square channel for the “Push Rod Tip” to slide through with an angled entrance to help align the machine every packet injection. A bolt hole was added in the center of

the clamp so that the “Push Rod Tip” impacts the metal bolt head instead of the plastic clamp to help reduce the wearing of the clamp. Small angled walls were added to the top of the back plate to help keep the packets from moving relative to the packet clamp once they are inside the clamp. The shape of the base of the packet clamp was changed to curved surfaces to match the shape of the stationary plate that they revolve around.

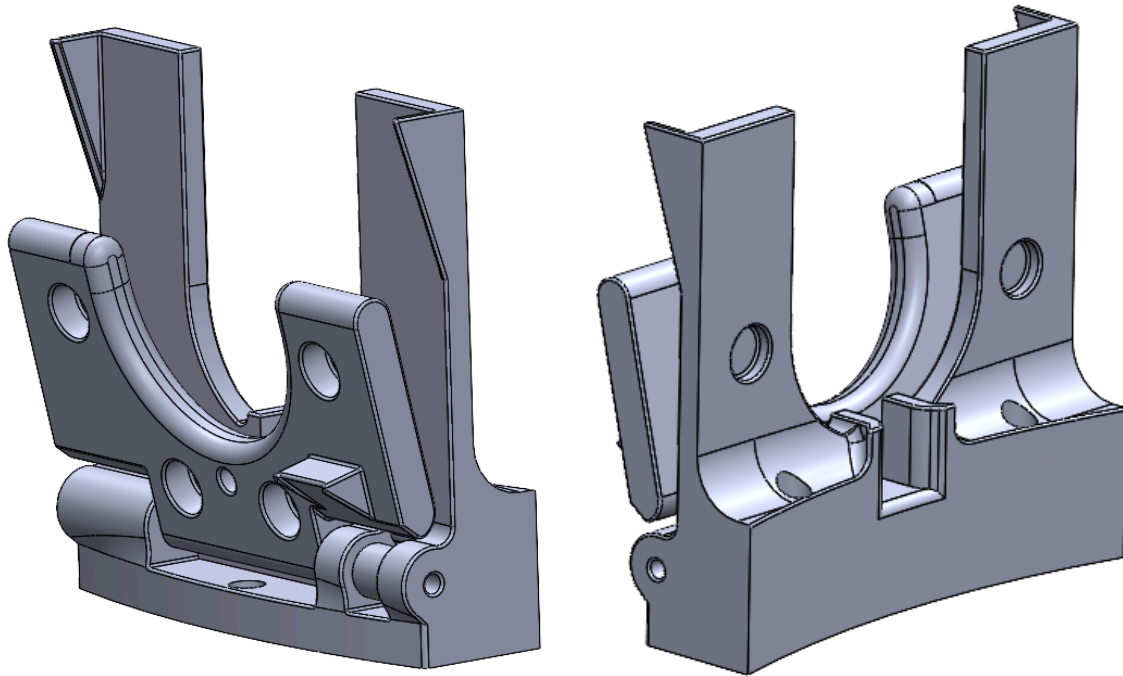


Figure 5: Final packet clamp design.

2.3.2 Stationary Structure Initial Design

The stationary structure is responsible for supporting almost every component of the packaging system, therefore it was critical to finalize the design early so that subsequent subsystems could be designed with size and spacing constraints known. The two main design types considered for the stationary structure were a fixed table and a floating table design. As suggested by the name, the difference between the two comes down to how the table is mounted in relation to the rest of the machine. For the fixed table design, the stationary table is directly connected to the main support column which also houses the rotating table. This design is more complex since the main column would have to support two tables at different levels while allowing the lower one to

freely rotate. In the floating table design, the stationary table is mounted directly to the table, over top of the rotating plate. This design would be simpler since each table would have its own support structure, however it would be much larger in footprint since the empty packets would have to be able to rotate underneath the table supports. Overall, the fixed type stationary table design was selected due to its ability to be compact as well as machined and assembled with increased accuracy. The initial design featured a center column with two turned steps. One would locate the rotating platform and the other the stationary table. To mount the rotating table, a turntable bearing would be used. This would allow the rotating table to be mounted below the stationary table, since the main support column could protrude through the center of the turntable bearing. This design can be seen in figure 6, below.

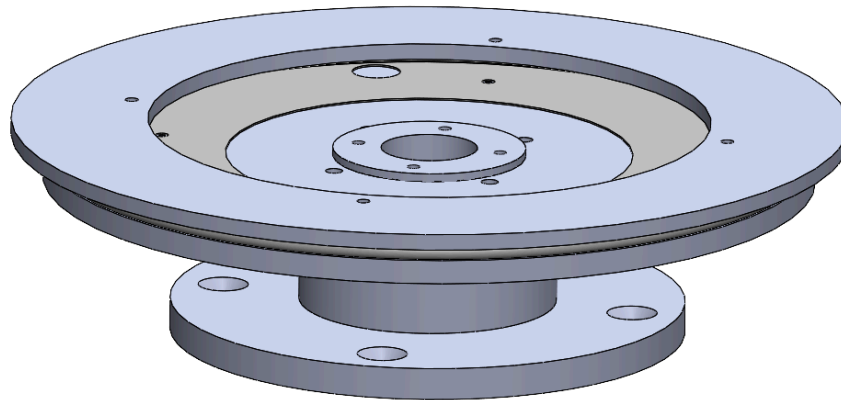


Figure 6: Turntable bearing base design with stationary plate removed for clarity.

2.3.3 Stationary Structure Final Design

Due to the cost of the large diameter aluminum bar stock required for this design, an even more compact version was developed using the same design ideas. The solution was replacing the turntable bearing with a roller bearing and a custom bearing carrier. This revised design can be seen in the figures below.

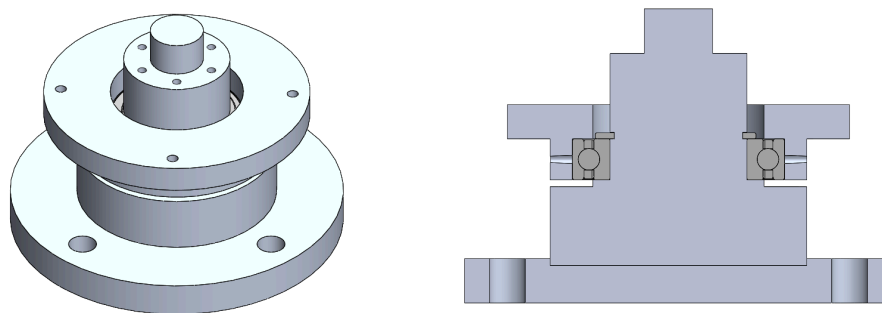


Figure 7: Final stationary structure and bearing carrier design. Stationary and rotating plates removed for clarity.

To reduce vibrations throughout the machine, it was determined that the baseplate was to be made out of .375" thick aluminum. This would also allow us to drill and tap the baseplate for simple mounting of each subsystem. Additionally, the stationary plate was designed as round so that the table itself wouldn't interfere with the packet's circular path and so that the subsystems could be laid out radially, allowing for multiple packets to be processed at once. The diameter of the plate was selected to be 400mm to allow for ample mounting space of all the required components. Unfortunately, this meant that the stationary plate had to be ordered from the E3 machine shop, since they have a larger waterjet cutter.

2.3.3 Rotating Plate Design

The overall design of the rotating plate didn't change much from ME481 other than its general shape. Once the stationary structure was finalized, the dimensions and final geometry of the rotating plate could be determined to fit within those constraints. The overall diameter of the rotating plate was selected as 460mm to ensure enough room for the packet clamps to be mounted to it and clear the stationary table. The center of the plate featured a clearance hole of 60mm as well as a bolt pattern that matched the bearing carrier. This would ensure that the rotary plate would fit overtop of the center support column, and allow it to be bolted to the bearing cup. Since the load on the rotating plate is only the weight of the packet clamps and the packets themselves, 0.125" thick aluminum sheet was used for the material to keep the weight and inertia of the rotating plate low. This would reduce the load on the stepper motor, allowing for quick packet transit times between substations. Additionally, because the rotating plate was so thin, the transmission system could fit in the narrow gap between the rotating and stationary plates. This

transmission system consisted of a NEMA 17 stepper motor mounted through the stationary plate and two gears. The pinion was mounted to the stepper motor with a custom machined coupler while the ring gear was bolted to the rotating plate with standoffs using dowel pins as an alignment tool to maintain concentricity, the design of which can be seen in figure 8 on the next page.

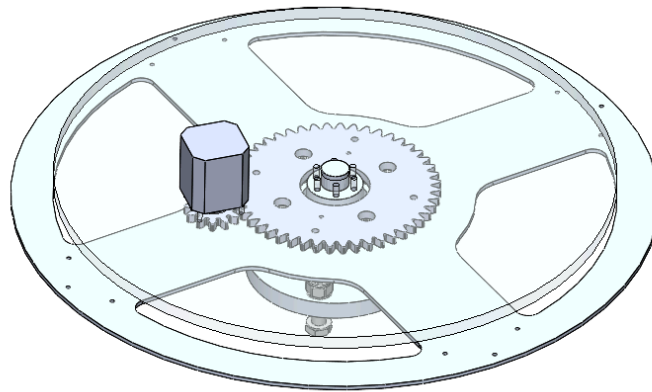


Figure 8: Rotating plate and transmission system.

A gear reduction of 4 to 1 was used to ensure that the stepper motor wouldn't be overloaded when accelerating and decelerating the rotating plate. Additionally it made the programming of the sub system convenient as one complete revolution of the stepper motor rotated the packet clamps exactly $\frac{1}{4}$ turn to the next subsystem. As with the stationary plate, the rotating plate had to be ordered from the E3 machine shop due to its size requirements. The overall structure of the complete rotating and stationary base can be seen in the figures below.

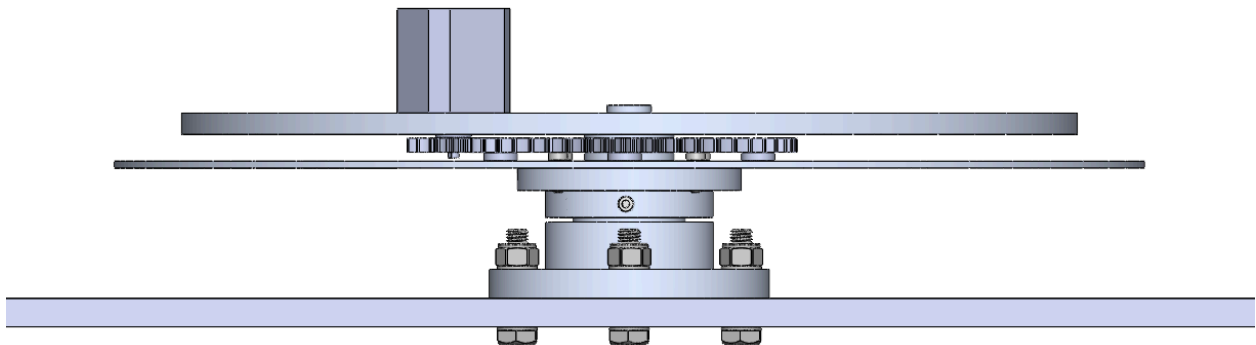


Figure 9: Side view of rotating base assembly.

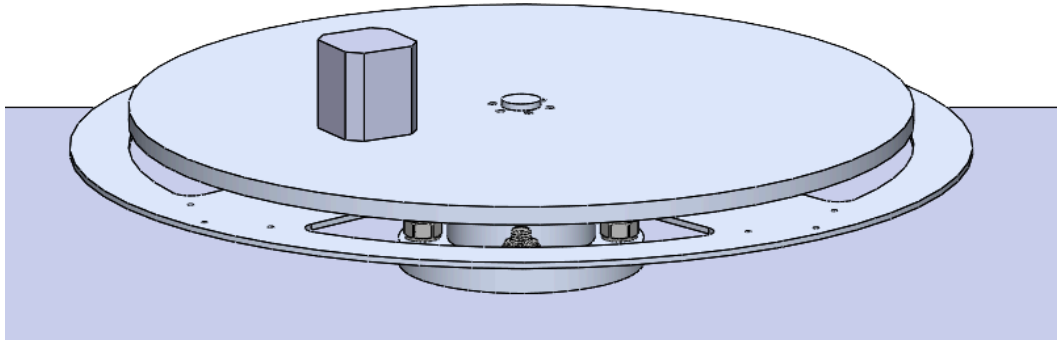


Figure 10: Isometric view of rotating base assembly.

2.4 packet Injection

The packet injection subsystem consisted of 2 main components: the packet feeder and the injection system. The packet feeder would hold bulk packets and present them to the injection system at a consistent height and location so that they could be pulled into the packet clamp one by one.

2.4.1 packet Feeder Initial Design

The initial design for the packet feeder at the end of the ME481 term was based on the proof of concept tests. It featured clear plastic sides as well as a thin front plate to match the test window the packets were pulled through in our verification. This design was simple with only a large spring and push plate mounted through the back of the feeder to keep the packet pushed to the front, as seen in figure 11 below.

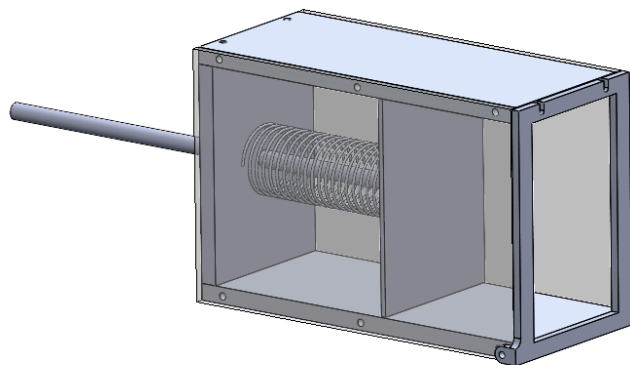


Figure 11: Initial design of packet feeder.

2.4.2 packet Feeder Final Design

The main concern with this design was overall lack of rigidity. This could cause the push rod to bind, preventing the packets from being kept at the front. To mitigate this, the design was modified to be completely made of aluminum to increase the rigidity of the structure.

Additionally, the push plate was mounted to a linear slide rail with the spring mounted below the floor to make the design more compact and reliable as a whole. Black UHMW plastic was overlaid on the bottom plate to reduce the friction between the housing and the packets. A 3d printed lid was made to flip open so that packets could be loaded from the top. These design changes can be seen in figure 12 below. The capacity of the packet feeder is approximately 225 empty packets.

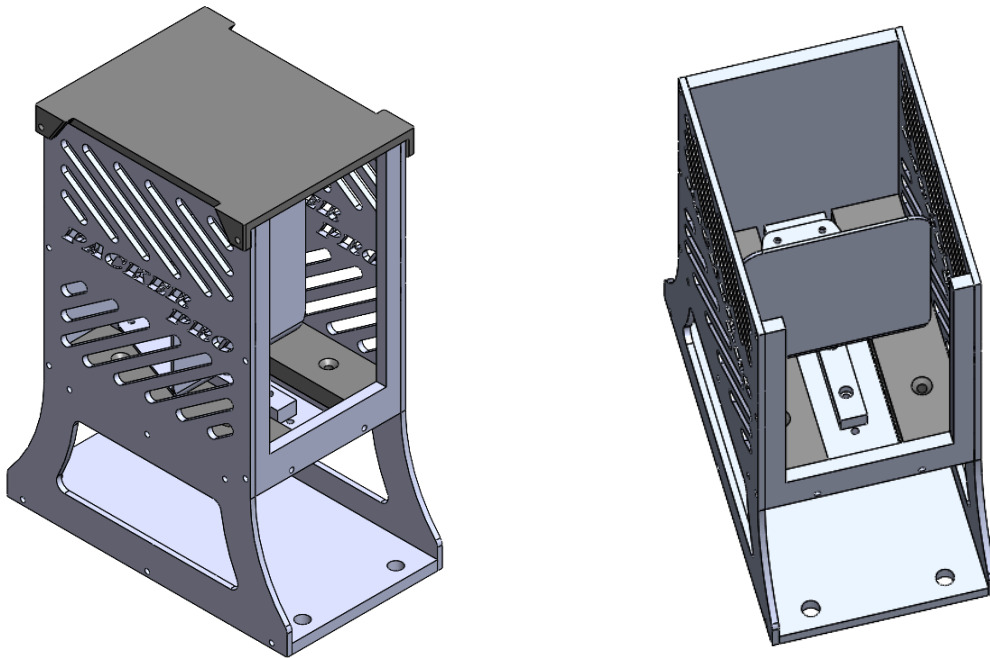


Figure 12: Final packet feeder design.

2.4.3 packet Injection System

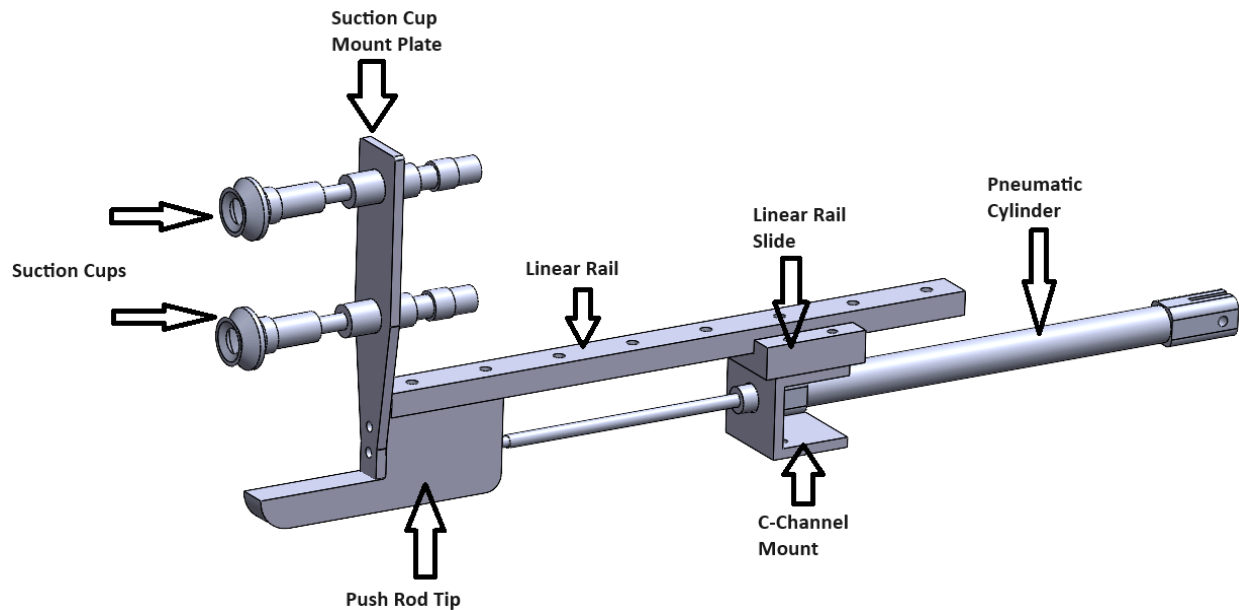


Figure 13: Final packet injection system design

The function of the “packet Injection System” is to transport a single packet from the “packet Feeder” into the “packet Clamp”. It does this by pushing a pair of suction cups into the stack of packets simultaneously the “Push Rod Tip” opens the “packet Clamp”. A vacuum is applied in the suction cups using a venturi valve to grab onto the packet. The cylinder can then retract and release the packet as the “Push Rod Tip” releases the “packet Clamp” allowing it to close on the packet holding it in place. The “packet Clamp” is an integral part of the “packet Injection System” because the “packet Clamp” is opened and closed by the same actuation that grabs the packet.

The “Push Rod Tip” is water jet out of $\frac{3}{8}$ ” thick aluminum plate then drilled and tapped by hand. A $\frac{1}{8}$ ” thick aluminum plate called the “Suction Cup Mount Plate” is bolted to the front of the “Push Rod Tip” and is what holds the suction cups in place. A tapped M4 hole is added into the rear of the “Push Rod Tip” to thread the pneumatic cylinder piston into. A linear rail is bolted through the top of the “Push Rod Tip” and secured at the rear with the “Linear Rail Slide” to allow for linear frictionless motion. A custom mount called the “C-Channel Mount” was machined from $\frac{1}{4}$ ” aluminum “c-channel ”; this component bolts to the “Linear Rail Slide” and

to the front of the pneumatic cylinder to fasten the whole assembly to the stationary plate using two M4 bolts through the bottom of the “C-Channel Mount”.

2.5 Seed Portioner Tower

The Seed portioner's function is to portion seeds from the hopper and transport a consistent volume of the seeds into a packet. The two inputs of this subsystem are bulk amounts of one seed manually placed in the hopper, and an unopened packet brought in by the rotary table.

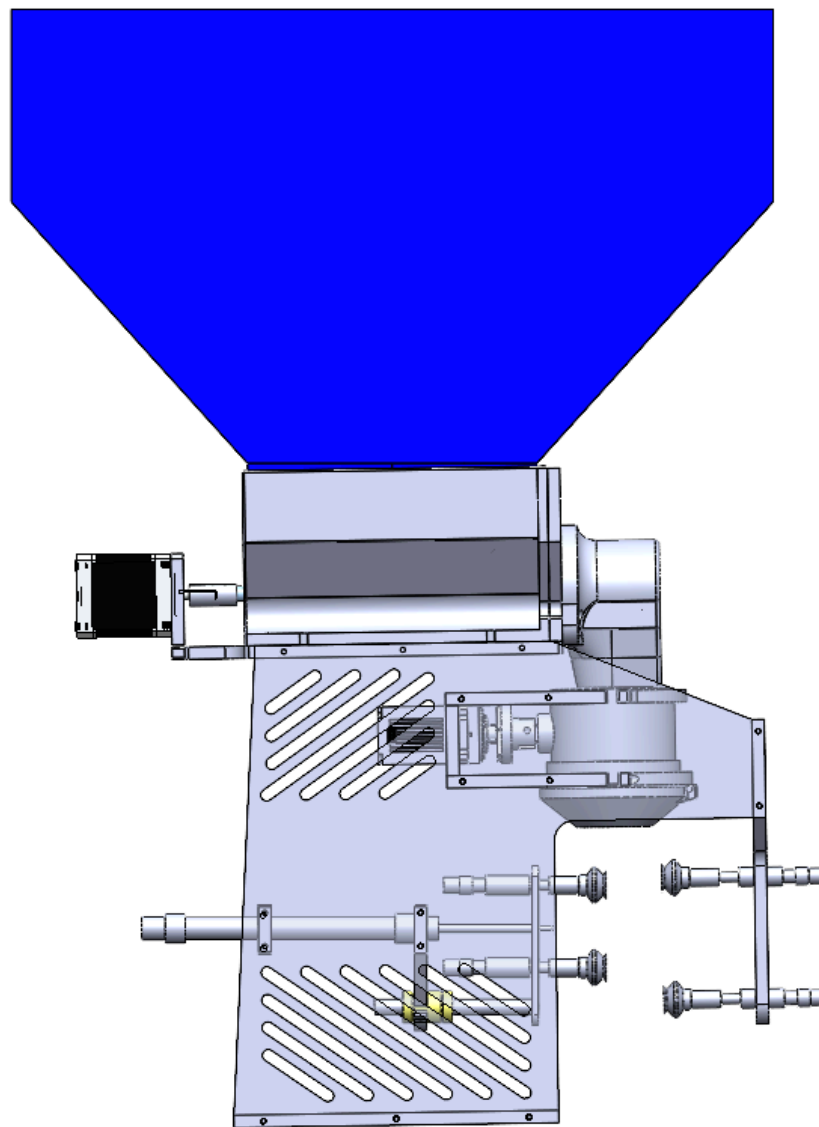


Figure 14: Final design of seed portioner tower

2.5.1.1 Seed Portioner Tower Frame

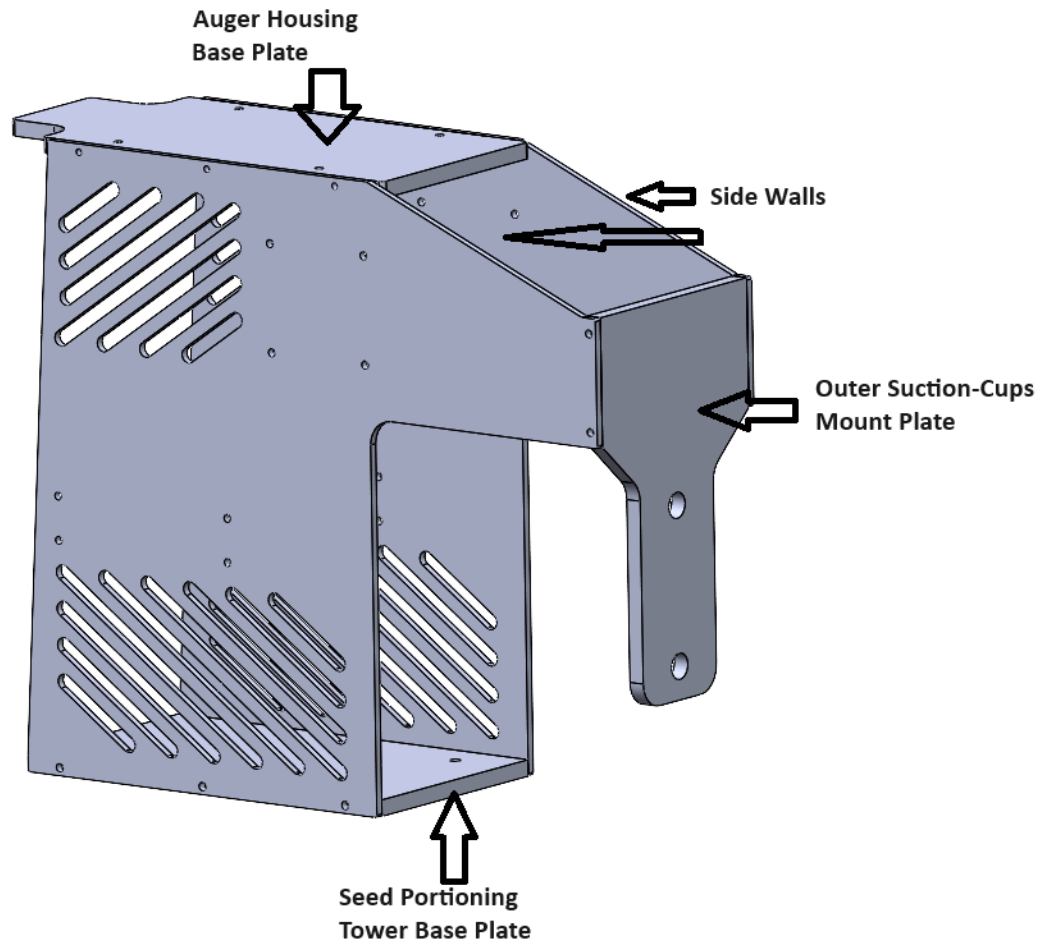


Figure 15: Final Seed Portioner Tower Frame

The “Seed portioning Tower Frame” is the structure that holds this entire subsystem together. The side walls are made from water jet 10 gauge aluminum sheet and are used to bolt any cross members together. The angled porting was cut out for light weighting and for aesthetics to better see the internals of the tower. The “seed Portioning Tower Base Plate” is a water jet ¼” plate with M6 bolt holes facing down to bolt the tower to the stationary plate.

2.5.1.1 Hopper and Auger Initial Design

. Originally we were planning to construct our own hopper out of 3D printed materials and fasten the panels together. This would enable us to have a custom interface between the hopper and the Auger housing. However we deemed this to be unnecessary and the extra design work

and manufacturing time outweighed the benefit of a more custom interface so we opted for an off the shelf auger.

In the initial design we had planned for an assembly made of 3D printed parts similar to the prototype shown below. This design worked well on a small scale but would be very difficult to make the interfaces of the different components strong enough for the larger final size we were planning for. This led to the decision of integrating some aluminum parts to strengthen the key areas of the auger housing and stepper motor mounts.



Figure 16: Initial Auger and Auger Housing design

A notable design issue with the first auger housing is the small gap for the seeds to fall through to fill around the auger shown in figure 17. This was initially done to better contain the auger by increasing the contact area between the auger and the auger housing. This flaw was especially bad for larger seed varieties. The initial design did not include bearings and relied on the friction fit between the auger and auger housing, we determined over long use these parts would wear out so we opted to include bearings in the final design.

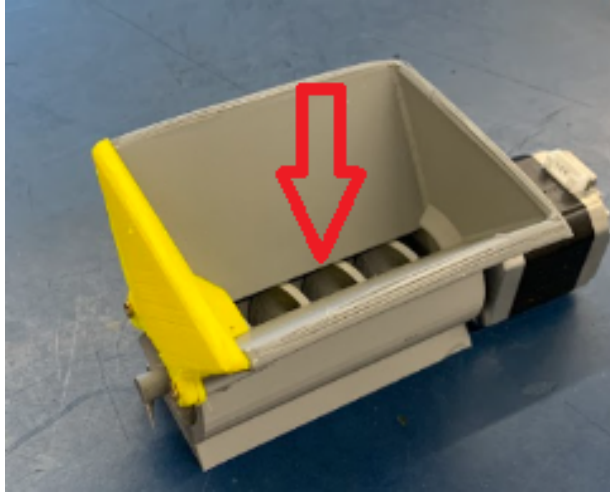


Figure 17: Auger housing small gap

2.5.1.2 Hopper and Auger Final Design

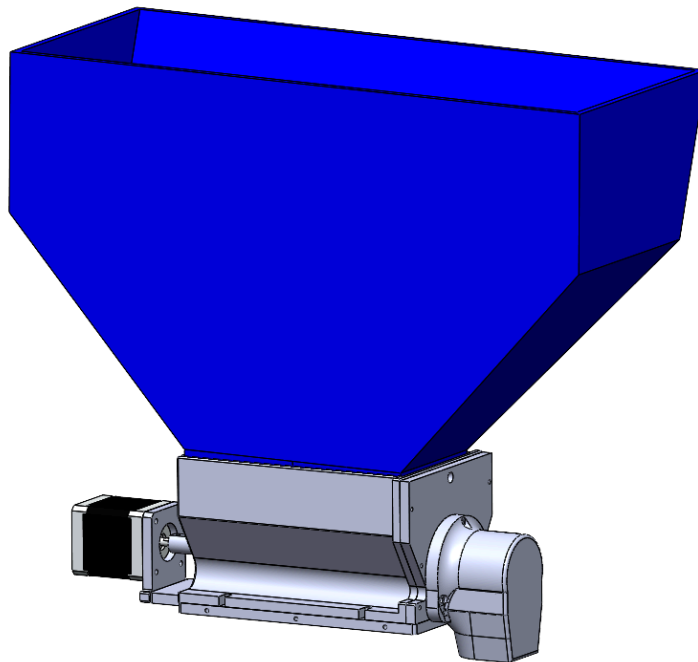


Figure 18: Hopper and auger final design

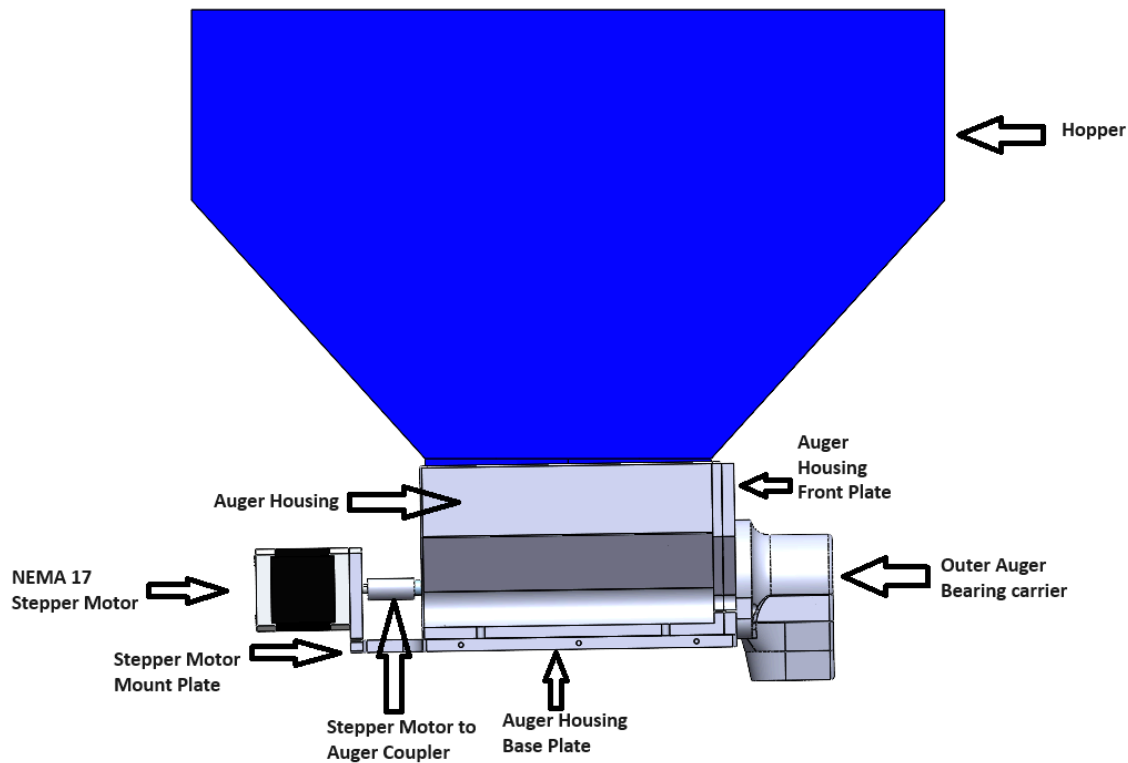


Figure 19: Hopper and auger final design labeled

The “Hopper” is a container that sits above the auger that holds onto the bulk seeds. Large quantities of seeds are manually poured into this hopper before operation. We decided to purchase one from an agricultural store that serves a very similar purpose for their customers. It was made of a very durable plastic and the spout fit with minimal design changes with our current design. This cost \$21.00 and saved the time of having to design and build our own. The “Auger Housing” is made of 3D printed materials however we opted to use a carbon fiber inlet filament to increase the strength.

To increase the strength of the interfaces between components we strategically added aluminum components. These aluminum components also contained threaded holes to bolt other components to which is a way more reliable and strong option compared to threaded inserts in 3D printed plastic that was previously considered. The aluminum components include the “Auger Housing Base Plate” which is water jet out of $\frac{1}{4}$ ” aluminum plate. This component is what fastens all the components to the side walls of the “Seed Portioning Tower”. The “Auger

Housing Front Plate” is also 1/4” waterjet aluminum with threaded M3 holes which connects the “Outer Auger Bearing Carrier” to the “Auger Housing”. The “Stepper Motor Mount Plate” is a water jet 1/4” aluminum plate and is what mounts the stepper motor in place, this plate is bolted to the side of the “Auger Housing Base Plate”.

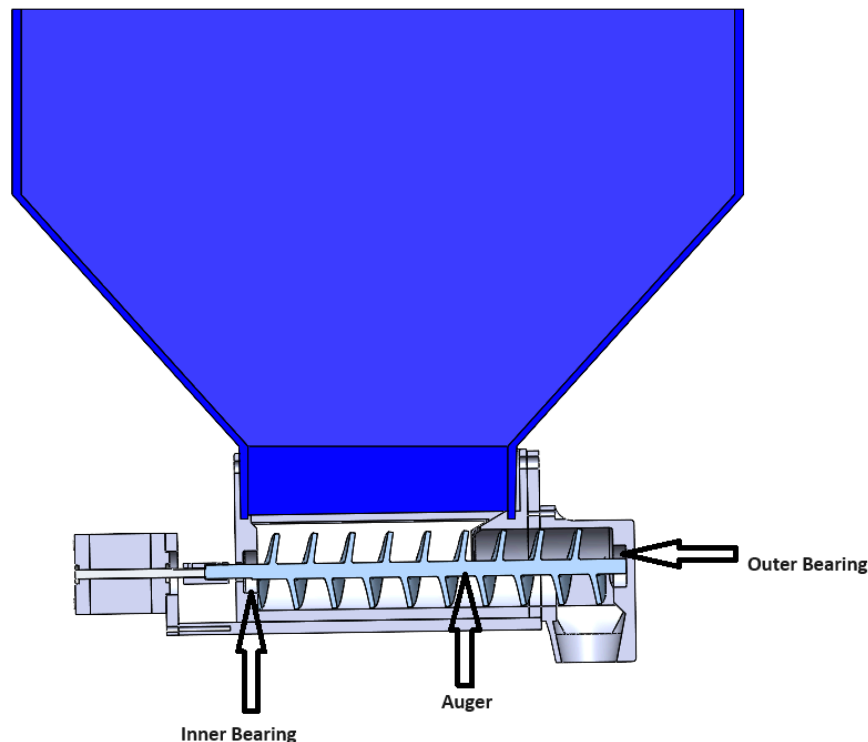


Figure 20: Auger and auger housing cross section labeled

The auger is what controls the flow and portioning of the seeds. It consists of a 3D printed auger using a carbon fiber inlaid filament for increased strength, it is mounted to a stepper motor via a custom aluminum coupler with set screws. The auger's diameter changed from 20mm to 30mm to allow for larger seeds and to move a larger volume of seeds per auger rotation. “The Stepper Motor to Auger Coupler” was turned in on a lathe out of aluminum then drilled and tapped to fit two aluminum set screws, one for the stepper motor shaft and one for the auger shaft. The auger is supported at both ends of its housing with ball bearings that are press fit into the Auger housing. These bearings allow for easy rotation of the auger and resist axial movement of the auger without greatly increasing friction.

2.5.2.1 Ball Valve and Spout Initial Design

At the end of 481 we did not plan on using a valve to control the flow of seeds. We simply had the auger direct the seeds to a spout that was integrated into the “Auger Housing” that would pour directly into the seed packages. A flaw with this spout design is it did not allow for a bearing to support the auger at the far end and would be held in place by the friction of the surfaces in contact with the auger. We believed this would cause too much wear on the auger and “Auger Housing”.

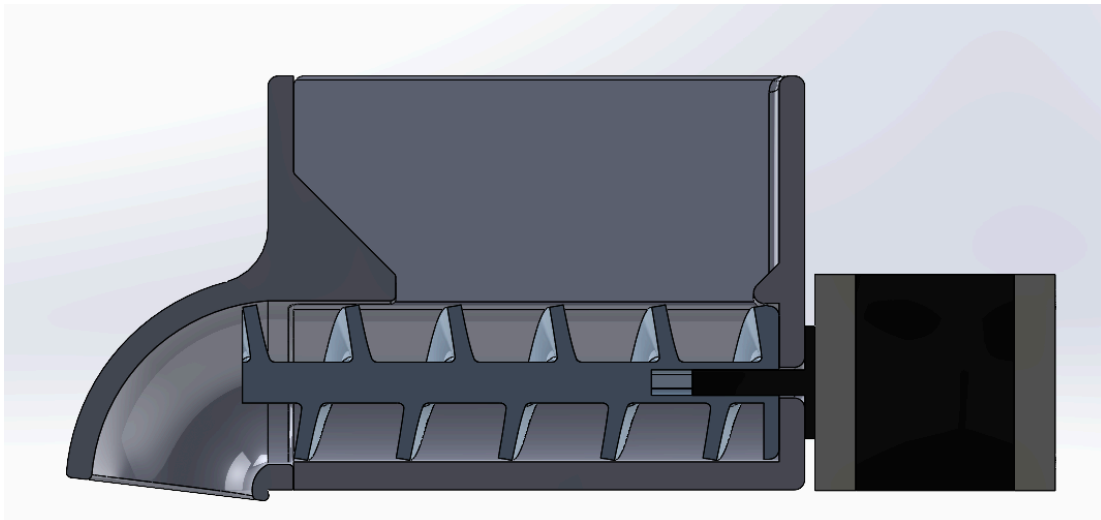


Figure 21: Auger and auger housing initial design

At the beginning of 482 to increase the production rate we decided to implement a ball valve to release the portioned amount of seeds all at once. This would allow us to rotate the auger before the packet is present and dump the seeds all at once as soon as the packet is spread, as opposed to waiting for the packet to be spread and rotating the auger. We considered two types of valves, a standard ball valve used for controlling flow of water and a gate valve. We went with the ball valve because through testing the valves with the flow of seeds we found the gate valve would get seeds caught in the mating surface of the valve and after repeated uses would prevent the valve from fully closing.

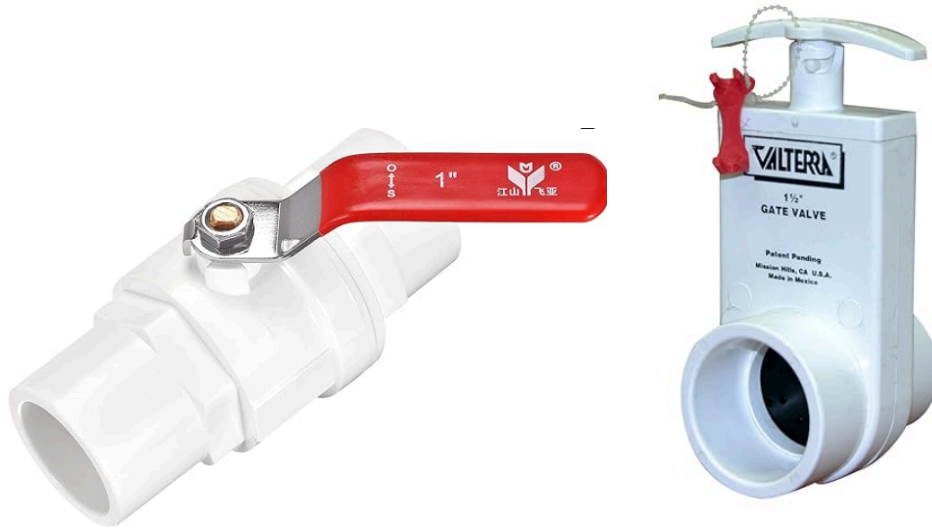


Figure 22: Ball valve used (left) and gate valve tested (right)

2.5.2.2 Ball Valve and Spout Final Design

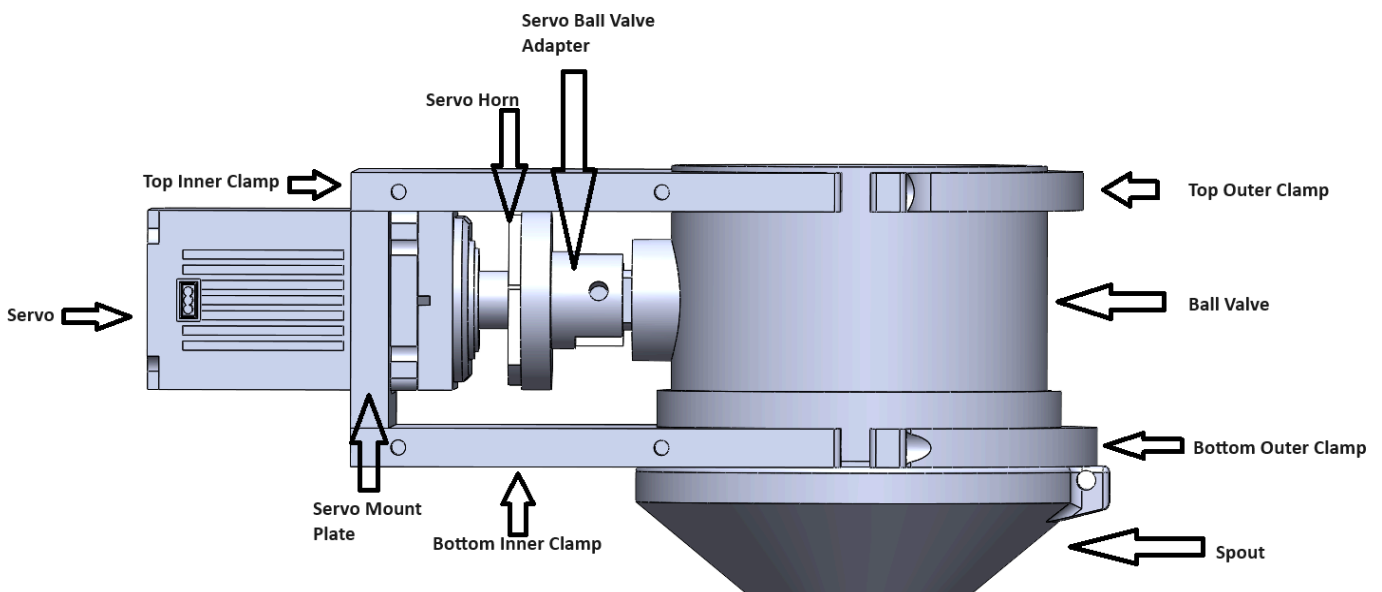


Figure 23: Ball valve and spout final design labeled

To minimize the space the ball valve occupied we cut off the plastic from both sides of the valve as much as we could without cutting into the internals of the valve. We used a servo from amazon to control the rotation of the ball valve, so we removed the original handle that came with the valve and designed a two piece coupler to mount to the old handle mount of the valve. The servo we selected could output 40 kg cm of torque and complete a 180 degree rotation in 0.3

seconds which was plenty fast and strong for this application. A “Servo Horn” also purchased from amazon fits onto the spline of the servo. We custom machined a “Servo Ball Valve Adapter” that bolts to the “Servo Horn” and slides over the rectangular shaft leaving the ball valve.

To fasten these components together and to mount to the side walls of the “Seed Portioning Tower” we designed an aluminum frame. This frame is water cut out of $\frac{1}{4}$ ” aluminum plate and then was drilled and tapped for M3 bolts. The “Top/Bottom Inner Mounts” have two tapped holes on either side to bolt into the “Side Walls”. There is a “Servo Mount Plate” that is bolted to the “Top/Bottom Inner Mounts” which fastens the servo in place using its existing 4 M3 bolt holes. This frame was built to not only support the valve internals but also act as a cross brace to increase the rigidity of the entire “Seed Portioning Tower”. To secure the ball valve to the frame we used clamps bolted to the “Top/Bottom Inner Mounts” that fit the circular profile of the valve. The “Spout” is used to direct the falling seeds into a more narrow stream to direct them more easily into the opened packet. This part was made of 3D printed PLA and is fastened to the bottom of the valve with a single M3 bolt and nut that clamps around the valve.

2.5.3 Packet Spreader Final Design

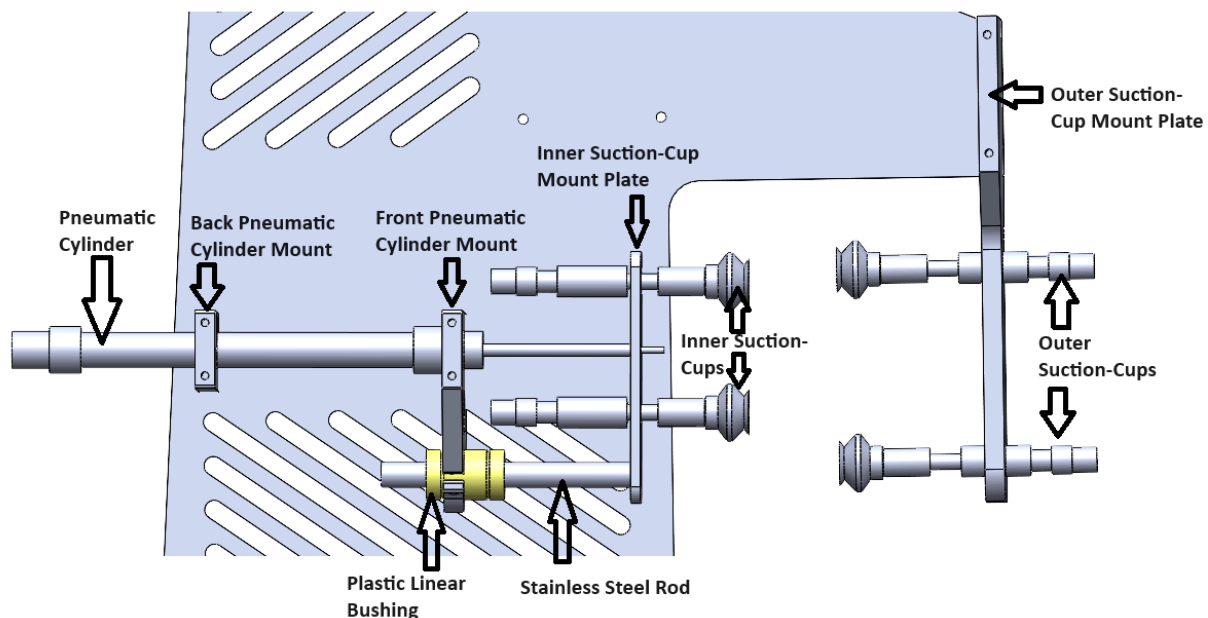


Figure 24: Packet spreader final design

In ME 481 the concept of using suction Cups to spread a packet open was validated however no design work was completed for this mechanism. The purpose of this mechanism is to spread the seed packets to allow them to be filled with seeds. This design consists of two stationary “Outer Suction-Cups” and two “Inner Suction-Cups” that move linearly to complete the spreading action. The two “Inner Suction-Cups” are translated using a pneumatic cylinder that pushes the inner suction cups into the outer suction cups, then the vacuum is generated in all four cups and the cylinder retracts, spreading the packet open. The vacuum is generated in the suction cups via two independent venturi valves. The venturi valves are plumbed so that the top and bottom suction cups use the same venturi valve. This was done so that if the bottom suction cups lose pressure which happens more frequently, the top suction cups which are more important do not have their suction impacted.

The “Inner Suction-Cup Mount Plate” is made from water jet $\frac{1}{8}$ ” aluminum sheet and is fasted to the pneumatic cylinder with a nut that threads onto the tip of the pneumatic cylinder. This component also holds both the “Inner Suction-Cups”. Similarly there is a “Outer Suction-Cup Mount Plate” made of $\frac{1}{4}$ ” aluminum plate, this holds the “Outer Suction-Cups” and also acts as a cross member for the entire structure. To prevent the rotation of the “Inner Suction-Cup Mount Plate” about the pneumatic cylinder an 8mm diameter stainless steel rod was attached to the bottom which fits inside a “Plastic Linear Bushing”. This allows for the linear movement of the “Inner suction-Cup Mount Plate” but restricts its rotation. The pneumatic cylinder is mounted at the front and back with a $\frac{1}{4}$ ” thick aluminum plate. The “Front Pneumatic Cylinder Mount” has two M3 bolt holes on either side to secure itself to the side walls. This mount also has a hole from the M12 threaded section of the pneumatic cylinder to fit through and is fastened with a M12 nut on the right side. On the bottom of the “Front Pneumatic Cylinder Mount” there is a clamp that fastens the “Plastic Linear Bushing”.

2.6 Heat Sealer

The heat sealer subsystem is one of the four main subsystems which had been determined previously in ME 481. As discussed previously, the main function of the heat sealer is to take an input of filled seed packets and apply an airtight seal by applying heat. Because this subsystem had not been verified in ME 481 the first step was to verify the subsystem could work.

2.6.1 Heat Sealer Proof of Concept

Conceptually, the heat sealer subsystem was envisioned as a mechanical press which would apply localized pressure to the top of filled seed packets, which were sandwiched between the press face and a resistive heat element. This process could be simulated with a generic hair curler by manually clamping filled packets in between the heat elements [Figure 25].



Figure 25: Generic hair curler used to test heat sealer subsystem

The main focus of the verification was twofold. At what temperature and for how long does heat need to be applied to the packets to apply airtight seals and will a press-style heat sealer leave an undesirable surface finish to the sealed packets. Thus, tests were conducted at various temperatures with packets at different fill volumes.

The results of these tests revealed an optimal sealing temperature of approximately 150 degrees celsius, applied for around 3 seconds [Figure 26]. Additionally, as long as packets were not filled to above approximately 60% of their 150 mL total interior volume, the heat sealer would seal them with an acceptable surface finish (without ripples or air pockets).



Figure 26: Acceptable seal at 150 degrees (left), unacceptable seal at 175 degrees (right)

2.6.2 Initial Design

The initial design of the heat sealer would take the main functions of the manual verification tests and attempt to apply them to a fully mechanical design which could be automated and integrated along with the other subsystems. The design consisted of large rectangular housing which provided support to a pneumatically actuated press and a heat element [Figure 27].

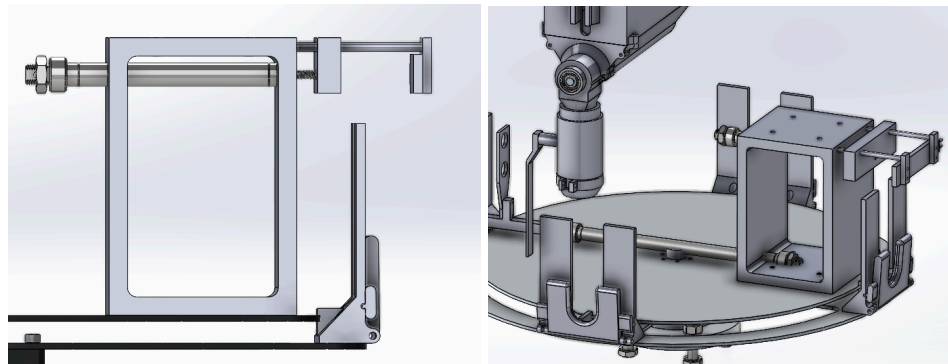


Figure 27: Initial heat sealer subsystem design

The main issues with this initial design were the inefficient use of material and space of the support structure and the effectiveness of the pneumatic press. Given the group's use of primarily machined aluminum, the rectangular support structure would use a large amount of expensive aluminum stock. While this would increase the overall strength of the structure, this was deemed unnecessary as the structure would not be supporting much weight and the chosen pneumatic cylinder would only be outputting around 25-30 N of force.

Additionally, the clamp would be supported by two cylindrical rods to counteract any bending moments from the pneumatic cylinder and to provide stability in the actuation. This would add resistance in the form of friction during actuation which could reduce reliability and increase wear on the subsystem.

2.6.3 Final Design

The subsequent iteration of the heat sealer subsystem aimed to address the issues with the initial design, along with considerations for simplifying manufacturing. The main support structure would now be comprised of four separate waterjet aluminum pieces which would be secured together with standard hex-head screws. Two triangular support columns provide the main support to the pneumatic cylinder. This construction results in a significant reduction in material and space usage [Figure 28].

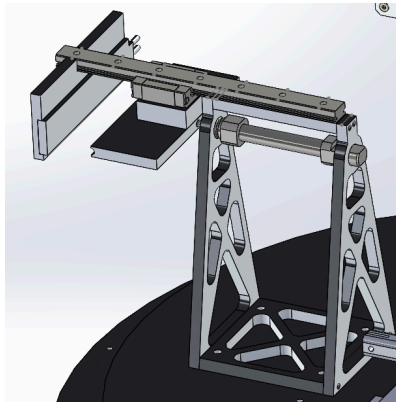


Figure 28: Final design of heat sealer subsystem

A linear rail along the top of the heat sealer also provided additional structural support as well as eliminating any concerns over the bending moments and added friction. The clamp would be secured directly to a slide mounted on the linear rail. Finally, the resistive heat element would be mounted to the end of the linear rail with an aluminum bracket. The below figure shows the final construction and assembly of the heat sealer subsystem.



Figure 29: Final assembly of heat sealer subsystem

2.7 packet Ejection

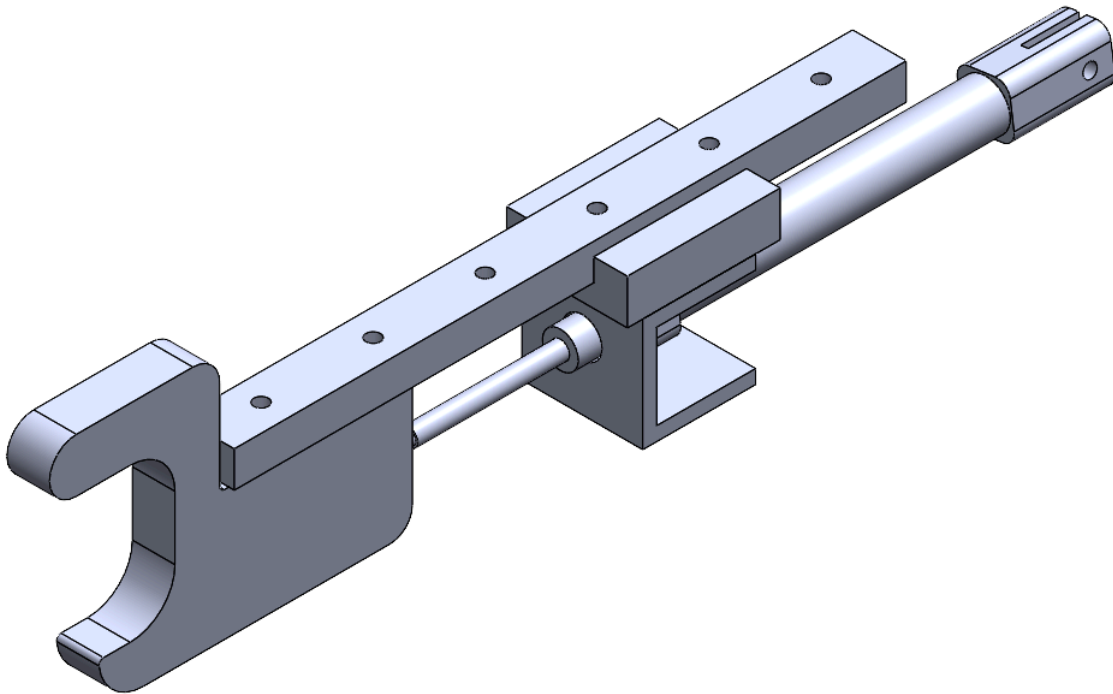


Figure 30: packet ejection final design

The packet ejection is a simple subsystem that does not require design iterations because the mechanics are very similar to the packet injection final design. The ejection tip is water jet out of $\frac{3}{8}$ " thick aluminum and has an identical profile to the "Push Rod Tip" used in the packet

injection. An additional arm is included above that to help angle the packet away from the clamp to ensure repeatable ejections. The subsystem is fastened to the stationary plate identically to the packet injection using another “C-Channel Mount” fastened down with M4 bolts.

2.8 Arduino Programs

2.8.1 Single packet Processing

The first program we made once the mechanical system was complete was a program that would fully process a single packet. The purpose of this program was to examine each subsystem working individually without all subsystems being used at once to make any issues more visible. It also served as a good demonstration tool during symposium to discuss what actions happen to each packet without the machine injecting more packets that would add confusion. The code was laid out as a bunch of functions being executed in a row and each function corresponds to a subsystem completing its entire process followed by a 90 degree rotation of the base.

2.8.2 Full Production

The full production program consists of four operation states, start up, steady state, last cycle, and emergency stop. Start up state occurs when the first 3 packets are being processed and the purpose is to not use the subsystems that don't have a packet at them yet. Once the fourth packet is injected, steady state can begin; this state is entered when there is a packet at every subsystem so that each subsystem can be functioning simultaneously. This state is exited and the program enters its last cycle when the current packet count is equal to the amount of packets that will be processed minus three. The last cycle state is essentially the opposite of start up, the program stops the subsystems in chronological order so that when packets stop being injected a subsystem does not attempt to do its operation without a packet present.

2.8.4 Tuning and Speed Adjustments

In order to tune the system delays are placed before and after each action and the variable determining the duration of the delay is set at the top of the program. Instead of using a simple delay function we made a custom delay function so that at every point in the program we can run checks to make sure an input to the serial monitor has not been entered. For example after we

tell the rotating base to turn 90 degrees we have a delay so the subsystems don't begin too early, if we used the premade delay function in arduino the program would not be able to read a serial input until the delay is completed which is not ideal for things like an Emergency stop that should respond immediately. The custom delay function checks the serial input constantly until the set delay time runs out. Then the program exits the custom delay function and moves onto the next step. A speed factor is multiplied to each delay value so that by changing the single speed factor variable the entire program can be slowed down.

2.8.3 Emergency Stop

As mentioned above the custom delay function checks for serial inputs at all times. We added a boolean “Stop” flag that gets turned too true if a ‘s’ is entered into the serial monitor. Flag check points are placed after each loop in the program so that if a ‘s’ was entered the program will exit to the outermost loop where a purge system function will be entered given that “stop” is true. The purge function ejects every packet in the system by rotating each packet to the ejection subsystem and running the standard ejection subsystem.

2.8 Electrical System

The complete system operation including all subsystems would need a robust and adaptable electrical system to provide the correct power to all electrical components. The system would need to provide power to the following components:

- Qty(2) - TB6600 Stepper motor drivers (rotating base and auger).
- Qty(2) - NEMA 17 Stepper motors (rotating base and auger).
- Qty(6) - Pneumatic solenoids (pneumatic cylinders and vacuum suction cups).
- Qty(2) - MOSFET boards (pneumatic cylinders and vacuum suction cups).
- Qty(1) - Step-up transformer (heat sealer).
- Qty(1) - Resistive heat element (heat sealer).
- Qty(1) - Step-down transformer (seed portioner).
- Qty(1) - Servo motor (seed portioner).
- Qty(1) - Arduino Mega.

2.8.1 System Structure

Rather than powering each of the above components separately, a single power supply for the entire system was desirable. The end user would simply connect a single power cord into a standard 110V AC power outlet and the power would be converted to DC and subsequently distributed to all the electrical components.

Thus, an external 12V DC power supply provides a constant voltage to a power distribution board which operates in parallel, supplying the same voltage to each of the components at varying currents [Figure 31]. Additionally, the system has a master switch to easily turn on or off the power in the case of an emergency.

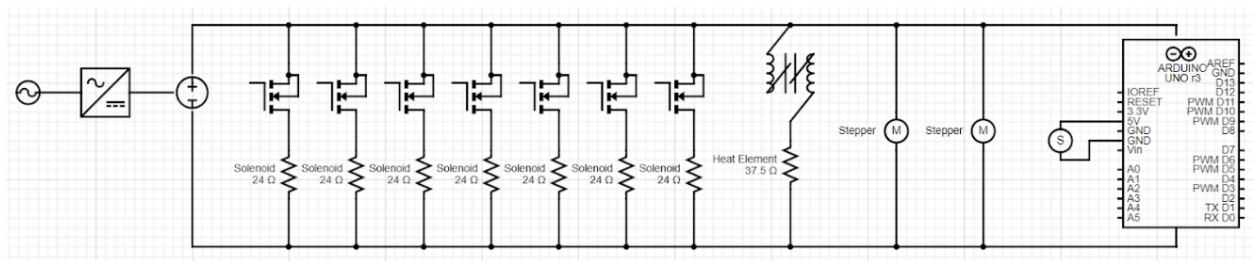


Figure 31: Initial electrical system schematic

Each parallel connection also has an added layer of safety from a user-replaceable fuse to ensure any surges in power draw do not result in catastrophic failure.

2.8.1.1 Stepper Motors

As discussed previously, stepper motors were to be used to rotate the rotating base subsystem and to actuate the auger in the seed portioning subsystem. Stepper motors require specialized PCBs known as drivers to function which take an input of power from 4-48V at varying amperage as well as a series of low-voltage signals. Given minimal torque requirements for both steppers, a relatively low standard 12V was selected. At 12V the steppers would draw a maximum of 1.9 Amps.

2.8.1.2 Pneumatics

To control the actuation of both the pneumatic cylinders and vacuum suction cups a series of pneumatic solenoids were required. Pneumatic solenoids take an input of compressed air and open or close their integrated valves depending on the signal they receive. The chosen pneumatic solenoids are normally closed and require a 12V signal to open. Given the Arduino microcontroller cannot output 12V signals, an external board was required.

Thus the chosen solution was to use a series of corresponding MOSFET gates which would act as an electrical relay. The MOSFETs would be given a constant supply of 12V power and would be normally closed. Once a 5V signal was received from the Arduino, the MOSFET would open and allow the 12V power to flow to the pneumatic solenoid which would complete the requested action (opening/closing a pneumatic cylinder or turning on/off a vacuum suction cup).

Given a total of six pneumatic solenoids, six corresponding MOSFETs were required, each drawing 12V at 0.5 Amps.

2.8.1.3 Heat Sealer

The resistive heat element is required to output a constant 70 Watts of heat in steady-state operation. By definition, resistive heaters operate at 100% efficiency as they convert all of the inputted electrical power into thermal heat. Thus, the heat element would need approximately 70 Watts of power in steady-state.

Given the selected resistive element's intrinsic resistance, the inputted power would be constrained by the voltage. This resistance is not constant and varies linearly with a change in temperature [2]. As a result, the heat element has an elevated resistance at room temperature which falls gradually as the temperature increases. Thus, to determine the optimal input power, tests were conducted by varying the voltage while maintaining a steady-state power draw of approximately 70 Watts. These tests revealed an ideal operating voltage of 35V (2 amps).

As a result, the heat element required a step-up transformer to convert the input voltage.

2.8.1.4 Servo Motor

Unlike stepper motors, servo motors do not require an external driver board to function and can be powered directly along with a 5V signal from a microcontroller. Servos generally operate at a set power in which the speed of actuation can be adjusted based on the inputted voltage. Testing the servo motor at different voltages revealed an optimal input voltage of around 6V. Similarly to the heat element, this meant the servo required a step-down transformer to convert the input voltage.

2.8.1.5 Arduino Microcontroller

To control all of the above electrical equipment, a single Arduino microcontroller was selected to simplify the process. Arduinos require a stable 12V input of power to function optimally.

2.8.2 System Assembly

In total the electrical system requires around 175 Watts of 12V power (see appendix section 10.2.1 for more details).

Most of the electrical components were to be located on a single electrical tower to minimize space usage and improved organization. The electrical tower thus contains the power distribution board, two MOSFET boards, two TB6600 stepper drivers, the 6V step-down transformer and the Arduino. These components would then be mounted vertically on a 3D-printed enclosure. To reduce any potential electrical hazards, the electrical distribution board and main power connection would be located in an enclosed region with a sliding acrylic shield [Figure 32].

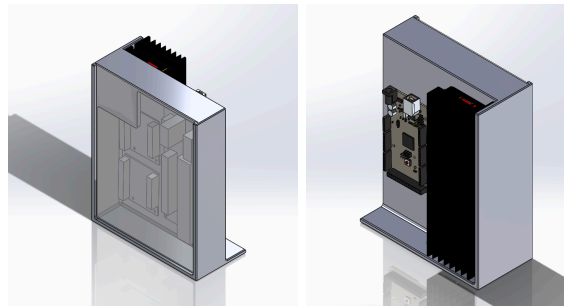


Figure 32: Final design of electrical tower

The user operates the electrical system by flipping the main electrical switch located on the top of the electrical tower. Furthermore, if adjustments need to be made or if any fuses have been blown, the user can easily access the fuses on the distribution board by removing the acrylic shield. The final assembly of the complete electrical tower can be seen in the figure below.

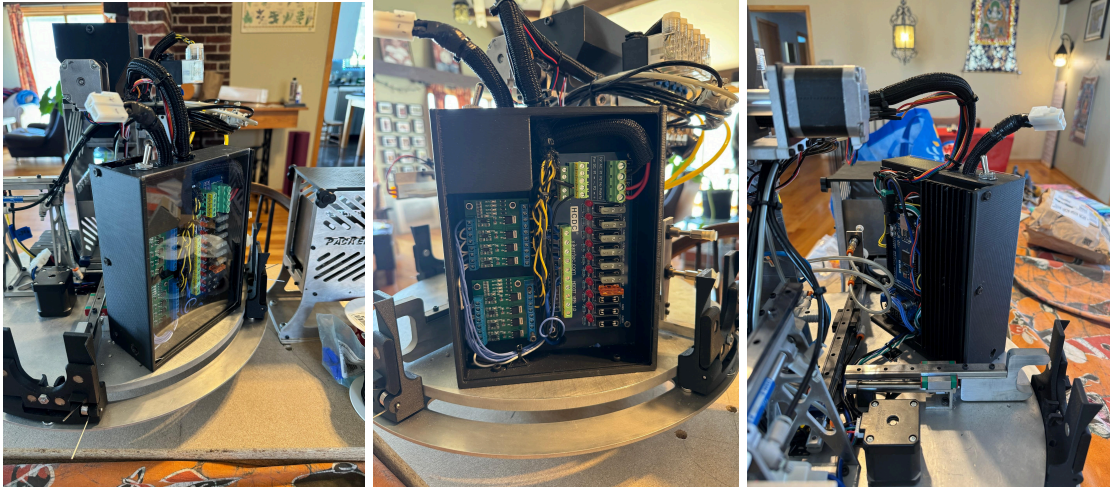


Figure 33: Final assembly of electrical tower

4.0 Design Verification of Engineering Specification

4.1 Heat Clamp Thermal Simulation

To ensure proper seals from the heat sealer subsystem, the resistive heating element would need to be at a consistent 150 degrees celsius in steady state operation. Given the surrounding structure and environment, the element would need to output a set heat flux. To determine the required heat flux a thermal CFD (Computational Fluid Dynamics) simulation was conducted.

The simulation used the solid model of the heat sealer subsystem assuming a natural convection fluid flow scheme. As a result the fluid domain was set at approximately five times the volume of the subsystem with zero mass flow at the domains. The initial conditions of the subsystem and surrounding air was set at room temperature. The exterior faces of the resistive heat element were set at 150 degrees celsius. The results of the simulation can be seen in the figure below.

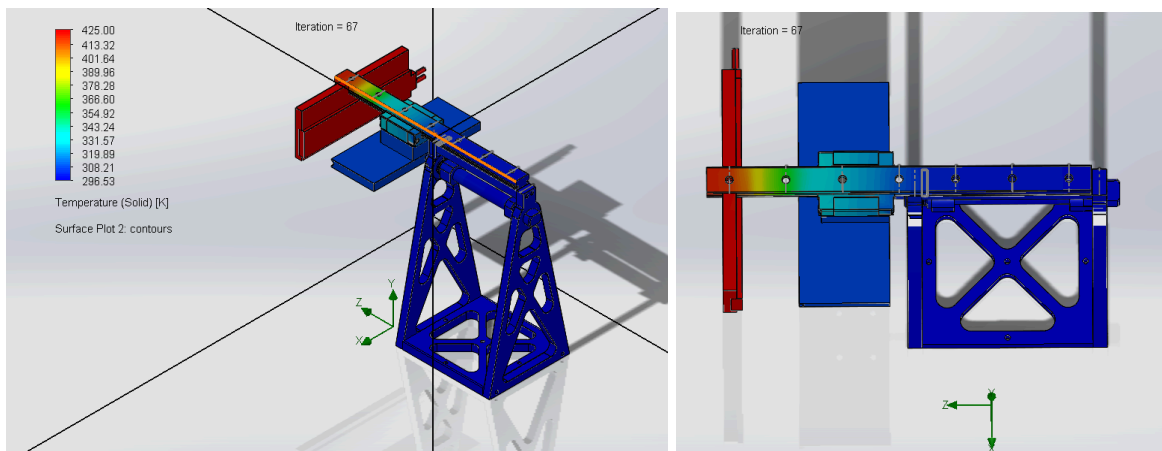


Figure 34: Thermal simulation of heat sealer subsystem in steady-state

Given these conditions at steady-state the heat element would be outputting approximately 70 Watts of heat.

4.2 Seed portioning Verification

To ensure accurate and consistent seed portioning we repeated our tests done in 481 where we rotate the auger a single rotation multiple times in a row and record the volume of seeds. The

final verification we completed was done using skittles and we recorded across 40 repetitions and calculated the standard deviation of the number of skittles.

Table 1: Seed portioner subsystem verification results

Number of Repetitions	Average # of Skittles	Standard Deviation	Range
40	0.7225	0.963	4

4.3 Production Rate Verification

The production rate was simply calculated by running the machine at steady state while recording the number of packets processed and tracking the time. Across a 20 packet test it took the machine 1 minute and 20 seconds resulting in a production rate of 1 packet every 4 seconds or 15 packets a minute.

5.0 Safety & Regulatory Considerations

5.1 Safety Concerns

Given one of the key design objectives being a user-friendly interface, safety was a primary concern in the design of the system. There were three main areas of concern.

5.1.1 Pinch Points Hazard

The first safety concern were the potential pinch points due to multiple moving parts amongst the subsystems. Sheets of acrylic were planned to be put in place as guards to minimize this hazard however, through additional testing these guards were deemed unnecessary. In practice, none of the moving parts are moving with enough force/velocity to be of concern. Additionally, the system is relatively compact and intuitive and is able to run without any direct manipulation of any of the mechanical parts.

5.1.2 Burning Hazard

The next safety concern was the potential burning hazard associated with the heat sealer subsystem. Given that the heat element's steady-state operating temperature is 150 degrees celsius, this presents a possible hazard to anybody directly touching the element. This hazard was deemed acceptable after analyzing the results of the heat simulation discussed previously in section 4.1. The steel and aluminum construction of the heat sealer subsystem dissipates heat at an elevated rate. As shown in figure 34, the results show a large delta in the temperature of the individual components of the heat sealing subsystem. From a maximum of 150 degrees celsius to a minimum of 23 degrees celsius, the only components which can potentially burn the user are the heat element itself and the surrounding clamp.

By adding a clear warning label to the hot components, the risks associated with this hazard are deemed acceptable.

5.1.3 Electrical Hazard

The final safety concern was the potential electrical hazard. Given the system utilizes standard 110V AC from standard wall outlets and draws between 100-200 Watts, there are electrical risks must be considered. The main electrical risk is evidently in converting the AC power from the wall to DC power. This risk was eliminated by using an off-the-shelf AC to DC power supply from a reputable brand. The power supply was also placed in a custom enclosure to further reduce any safety hazards. This power supply converted the 110V AC power to safer 12V DC. At this low voltage, bare skin offers adequate resistance thus electricity will not pass through someone who touches any of the onboard electrical components [3]. However, to add an additional level of safety, the onboard electrical tower includes a protective acrylic cover which covers the power distribution board and the main electrical connection.

5.1.4 Emergency Shutdown

In the event of injury or catastrophic failure, PackerPro incorporates emergency shutdown procedures. Given an emergency scenario, the user can opt to shutdown the system regardless of its current state depending on the severity of the situation. If the shutdown request is purely due to malfunction and there is no risk of injury, the user can press “e” to initiate the shutdown program. As discussed previously, this program will stop any current subsystem process and immediately begin to eject all packets, ending in a ready-state.

If the shutdown request is more severe - due to injury or potential injury - the user can immediately turn off all electrical power to the system by flipping the master electrical switch. Without electrical power all subsystems are rendered harmless.

5.2 Regulatory Considerations

Another critical area of consideration in this project was the regulatory considerations associated with handling seeds. Packaged seeds are regulated under the CFIA (Canadian Food Inspection Agency) in Canada and equivalent federal agencies globally. The CFIA outlines requirements for minimizing contamination when packaging seeds for sale in Canada. Given the stated requirement of the system to be able to operate many different varieties of seeds, the machine

must thus have a system to decontaminate any surfaces which may come in contact with seeds when switching to a different variety.

The solution to this is twofold. First, the main compressed air connection is of a quick disconnect variety and can be easily connected/disconnected. During operation the user is expected to disconnect the compressed air connection after a batch of seed packets has been completed. The user will then connect the compressed air connection to a compressed air blow nozzle and will blow down all the exposed surfaces, removing all larger contaminants.

Additionally, all components which come in contact with seeds (auger, auger housing, hopper, valve, etc) are designed to be easily accessible and removable. For seeds with an increased cleanliness requirement, these components are to be removed by the user and cleaned manually in addition to the initial blowdown.

6.0 Project Management

6.1 Team Communication

Due to the small team size, verbal, in-person communication proved to be the most effective way to collaborate on the project. It was found early on in the project that this way of communicating led to efficient design critiquing and idea sharing since any questions could be answered on the spot with minimal misunderstanding. These in person conversations worked well since each term all of the team members had at least one class together, leading to impromptu meetings whenever required and preventing any wasted time that would occur waiting for a meeting to address a design issue.

If communication was necessary outside of regular hours or when any group member wasn't present, a group chat was used for quick messaging and planning. Additionally, in moments of greater urgency, phone calls between group members were allowed at any time of day or night.

6.2 Document and CAD File - Sharing and Storing

To effectively collaborate throughout the project, a shared Google Drive folder was used to store and work on all deliverable documents as well as share CAD files. To reduce the risk of using non-current CAD files throughout the design process, clear communication around model updates was used and each time a model was updated it would be put in a new folder labeled with its revision number. One team member would then ensure that the master model's parts were kept up to date with their latest revisions.

6.3 Deliverables and project timeline

At the beginning of each term, the team met and outlined the main deliverables that would be required for successful and continual progress to be made on the project. These included the main course deliverables, such as the initial, midterm and final design reviews, but also incorporated major project milestones that kept the group on track to complete the project by the end of the ME482 term. The complete list of all course deliverables as well as both the planned project milestones and actual milestone completion list can be found in section 10.1.1 of the

Appendix. One crucial feature of the original milestone plan was the inclusion of a three week buffer time at the end of the ME482 term. This was added in as part of the mitigation strategy to avoid the risk of project incompleteness, as highlighted by the risk register in section 10.1.5 of the Appendix. This buffer time meant that if everything went according to plan, the team would have the seed packaging system completed 3 weeks ahead of the symposium. In the event of any delays or issues, we would then still have 2 weeks of extra time to sort those issues out. As shown in the actual milestone completion list, these delays did happen as a result of waiting on components to be machined or shipped, resulting in the group working until the last week to finish the project. However due to the extra time in the schedule, this wasn't an issue and the symposium was a success.

6.4 Budget

6.4.1 In-Kind Contribution

There are three parties to consider when accounting for labor costs. The main contributor is the group's labor. Considering 780 hours in total split between two terms at \$28.50/hour (current co-op salary earnings average between work terms 5 and 6) the total is \$22 230. The next contributors are the faculty advisors. Assuming around 40 hours split between both terms and professors (hourly rate in appendix section 4.2), the total is \$3560. The final contributor is the client. Assuming a total of 30 hours split between both terms at an average of \$60/hour, the total is \$1800.

Adding all of these contributions together, the total in-kind contribution is \$27 590. Total time expenditure from group members roughly equated to the predicted 780 hours with a weekly average of 10 hours per week, per member across both terms.

$$10 \text{ [hours/(week*member)]} \times 28 \text{ [weeks]} \times 3 \text{ [members]} = 840 \text{ Hours}$$

Group work was distributed equally amongst team members with all contributing towards the design, manufacturing, construction, programming and testing of the system.

6.4.2 Total Materials Budget

In the ME481 term of the project, the projected spending for all necessary components and materials was roughly \$950. Throughout the ME482 term, it became apparent that we were going to overrun the budget. This was largely due to an underestimation of the cost of control components as well as the extra cost of getting the stationary and rotating plates made at the E3 machine shop. A detailed budget breakdown is included in section 10.1.4 of the Appendix. In short, the budget overrun was approximately 43%, as shown in the following breakdown.

Material Cost	=	\$380.92	
3D Print Cost	=	\$38.19	
Controls Component Cost	=	\$939.08	
Total Actual Cost	=	\$1358.19	
Predicted total budget	=	\$950	
Overrun	=	$(1358.19 - 950) / 950$	= 43 %

7.0 Team Lessons Learned

The largest lesson learned from the work completed in 481 is to avoid being stuck in a design loop without providing yourself additional information from testing and verification. We found ourselves doing too many design iterations based solely on intuition which lead us to not consider many little things that greatly impact the overall design. These issues would be addressed a lot quicker if we expedited each design iteration and jumped to testing right away.

We found our small group size to be an unexpected advantage. Previously we have worked in larger groups where communication with a large group can be more work than the actual work itself. In our small group of three we found it very efficient to catch everyone up to speed which saved a lot of time. It also put our group in a different mindset than if we were in a larger group, where each member took full accountability for things they were working on whereas in a large group it's easy to put things off under the assumption someone else will do it.

A technical lesson learned that we have also experienced before is the classic “measure twice cut once”. A couple times throughout the project we made costly mistakes that would have been avoided if we just measured an extra time before building the part.

8.0 Conclusion

In conclusion, this report aimed to summarize the work completed in the development, design, manufacturing, and testing of PackerPro. Beginning in ME 481, the group worked directly with Richters Herbs to produce a product which would automate their seed packing process. Specifically, a need exists to process bulk seeds into packages of smaller quantities, with minimal manual interaction.

PackerPro's solution was thus to incorporate a series of five main mechanical subsystems: packet injection, rotating base, seed portioner, heat sealer and packet ejection. In addition to the five main mechanical systems, a pneumatic circuit as well as an electrical system were required.

The final result was a self-contained machine which completed most of its stated objectives and is easily adaptable and intuitive for the end user. By simply loading PackerPro with seeds and packets, supplying a compressed air and electrical connection, the user can produce any number of filled packets (upper limit of 150~200) with the push of a button.

The group was recognized with the best prototype award and nominated for best overall award for their efforts and accomplishments.

8.1 Stakeholder Feedback and Next Steps

Proceeding the completion of PackerPro and the final demonstration of the product at the symposium, Richters Herbs and the group have outlined areas of interest for future development. As stated previously, PackerPro functions according to the stated specifications however Richters has identified specific items they would like to see improved prior to using the machine for full-scale seed production.

Beginning with the seed portioner subsystem, the auger should be manufactured with increased precision and strength. In its current form the auger can occasionally get stuck due to gaps between the threads and the auger housing. This issue becomes increasingly prevalent as the size of the processed seeds decreases. Manufacturing new augers of varying pitches out of different materials such as steel/aluminum or other high strength plastics is desirable.

Additionally, the ball valve in the seed portioner subsystem should be refined or replaced to address failures and increase reliability. Contaminants can enter the ball valve which in turn increases the friction during normal operation. Replacing the current servo motor with a more powerful motor or replacing the ball valve entirely with an alternative such as the originally considered gate valve may address this issue.

For the heat sealer subsystem, a minor improvement outlined by the group would be to incorporate a temperature sensor on the heat element. This sensor would actively check the local temperature to ensure the heat sealer is operating at a sufficient temperature to properly seal packets.

Another minor improvement outlined by the group is to improve the mounting plate which currently secures the rotating base and packet injection subsystems in place. Less deflection and movement in these mounting points could potentially increase the reliability of the packet injection process.

Finally, a failure detection system should be added to PackerPro. The system should include a series of active sensors (possibly lasers/IR detectors) which can indicate to the system internally if a packet has been improperly processed by one of the subsystems. For example, if the packet injection system fails to properly inject a packet, the system should detect this failure and proceed to not activate the remaining subsystems for this individual packet. The empty packet should be ejected from the machine and the system should not count the respective packet in its completed packet count.

9.0 List of References

[1] - Group 22, “Final Design Report: Seed Packer,” University of Waterloo, Waterloo, ON, Canada, 08, 2023

[2] - Ling, S. J. (2024, March 26). *8.1: Faraday and Lenz's laws*. Physics LibreTexts.
[https://phys.libretexts.org/Courses/Georgia_State_University/GSU-TM-Physics_II_\(2212\)/08%3A_A_Electromagnetic_Induction/8.01%3A_Faraday_and_Lenzs_Laws](https://phys.libretexts.org/Courses/Georgia_State_University/GSU-TM-Physics_II_(2212)/08%3A_A_Electromagnetic_Induction/8.01%3A_Faraday_and_Lenzs_Laws)

[3] - Fish, R. M., & Geddes, L. A. (2009, October 12). *Conduction of electrical current to and through the human body: A Review*. Eplasty.
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10.0 Appendices

10.1 Project Management

10.1.1 Project Objective and Deliverables

Spring 2023 ME481 Deliverables Schedule

W2	-	Project registration
W4	-	Finalize outline of required subsystems
W4	-	Design proposal report due
W5	-	First teamwork assessment due
W5	-	Complete concepting phase
W5	-	Concept evaluation meeting
W7	-	Midterm design presentation
W9	-	Finalize preliminary design
W10	-	Finalize outline of required controls systems
W11	-	Final design presentation
W12	-	Final design report due
W12	-	Submit final design report to Richters Herbs for review
W13	-	Second teamwork assessment due

Winter 2024 ME42 Deliverables Schedule

W1	-	Obtain feedback from Richters Herbs
W2	-	Initial design review
W4	-	Complete design revisions based on feedback
W6	-	Midterm design review
W7	-	Third teamwork assessment due
W9	-	Finish manufacturing of system
W11	-	Finish integration and testing of controls
W11	-	Final design review
W12	-	Public exhibition
W13	-	Final Report Due
W13	-	Fourth teamwork assessment

Spring 2023 ME481 Milestone Schedule - Planned schedule had no delays

- W4 - Outline all required subsystems and requirements
- W5 - Complete concepting phase. Evaluation meeting
- W7 - Continue preliminary design.
- W9 - Finalize preliminary design.
- W10 - Finalize preliminary controls outline.
- W11 - Begin work summary.
- W12 - Complete work summary.
- W13 - Submit a final report to all stakeholders.

Winter 2024 ME482 Milestone Schedule - PLANNED SCHEDULE

- W1 - Team meeting (recap progress to date & assign tasks/priorities)
- W2 - Finalize design for rotating base, heat clamp, and packet injection
- Order components and begin machining/3d printing
- W3 - Finish machining
- Build and verify rotary table and packet clamp
- Finalize design of packet feeder and seed portioner subassemblies
- Complete controls outline
- W4 - Order controls & pneumatic components
- Manufacture packet feeder and seed portioner subassemblies
- W5 - Begin integrating electrical and pneumatic actuators
- Verify ability to control system components
- W6 - Complete construction of system
- W7 - Program system operation
- Test seed dispensing and subsystem timing
- W8 - Finalize control settings for different seed types
- Test dispensing accuracy and adjust as necessary
- Make adjustments to reduce cycle time
- W9 - Extra time for delays
- W10 - Extra time for delays
- W11 - Symposium

Winter 2024 ME482 Milestone Schedule - ACTUAL SCHEDULE

W1	-	Team meeting (recap progress to date & assign tasks/priorities)
W2	-	Finalize design for rotating base, heat clamp, and packet injection
	-	Order components and begin machining/3d printing
W5	-	Finalize design of packet feeder and seed portioner subassemblies
W6	-	Complete controls outline
W7	-	Order controls & pneumatic components
W8	-	Verify ability to control system components
W9	-	Finish manufacturing all components
W10	-	Complete construction of system
	-	Integrate electrical and pneumatic actuators
	-	Program system operation
W11	-	Test seed dispensing and subsystem timing
	-	Make adjustments to reduce cycle time
	-	Symposium

10.1.2 Work/time Budgeted (In-Kind Budget Calculation)

University of Waterloo Co-op Mechanical Engineering Salary Data:

5th term average = \$27.75/hour

6th term average = \$29.40/hour

Combined average = $(27.75 + 29.40) \div 2 = \$28.50/\text{hour}$

Cost = \$28.50/hour × 780 hours = \$22 230

University of Waterloo Salary Data:

Prof. Mohammed Nassar: \$175 000 annual salary = \$84/hour

Prof. HJ Kwon: \$195 000 annual salary = \$94/hour

Cost = \$84/hour × 20 hours + \$94/hour × 20 hours = \$3560

Client Salary Estimate:

Assume \$60/hour

Cost = \$60/hour × 30 hours = \$1800

Total In-Kind Cost = \$27 590

10.1.4 Budget

10.1.4.1 Material

Order 1	Tuesday Jan 16, 2024			
Paid by Tyson via Watcard				
Description	Quantity	Unit Price	Measure	Total
Alum Solid Round 3 - 6061	0.420	\$71.60	foot	\$30.07
Alum Solid Round 5 - 6061	0.060	\$190.32	foot	\$11.42
Bolt Steel Hex GR 5 NC 1/2x1 3/4	1.000	\$0.41	each	\$0.41
			Sub Total	\$41.90
			Tax (13%)	\$5.45
			Total	\$47.35
Order 2	Tuesday Jan 30, 2024			
Paid by Tyson via Watcard				
Description	Quantity	Unit Price	Measure	Total
Alum Flat 1/4x6 - 6061	2.000	\$10.92	Foot	\$21.84
Alum Plate 1/4 - 6061 T6	1.000	\$30.16	Square Foot	\$30.16
Alum Sheet - 5052 H32 1/8 - .125	2.000	\$11.17	Square Foot	\$22.34
M3x0.5 SHCS 12mm lg.	40.000	\$0.03	Each	\$1.20
M4x0.7 SHCS 10mm lg.	2.000	\$0.04	Each	\$0.08
M4x0.7 SHCS 16mm lg.	6.000	\$0.08	Each	\$0.48
M4x0.7 SHCS 25mm lg.	2.000	\$0.11	Each	\$0.22
M5x0.8 SHCS 16mm lg.	8.000	\$0.08	Each	\$0.64
M10x1.5 SHCS 40mm lg.	4.000	\$0.79	Each	\$3.16
M4 Flat Washer	4.000	\$0.01	Each	\$0.04
M10 Flat Washer	8.000	\$0.11	Each	\$0.88
M4 Spring Lock Washer	2.000	\$0.02	Each	\$0.04
M10 Spring Lock Washer	4.000	\$0.09	Each	\$0.36
M4x0.7 Hex Nut	2.000	\$0.02	Each	\$0.04
M10x1.5 Hex Nut	4.000	\$0.08	Each	\$0.32
			Sub Total	\$81.80
			Tax (13%)	\$10.63
			Total	\$92.43
Order 3	Wednesday Feb 21, 2024			
Paid by Justin via Watcard				
Description	Quantity	Unit Price	Measure	Total
Alum Solid Round 1 - 6061	1.000	\$5.91	foot	\$5.91
Alum Angles 6061 1.5 x .25	1.000	\$4.57	foot	\$4.57
			Sub Total	\$10.48
			Tax (13%)	\$1.36
			Total	\$11.84

Order 4		Friday Feb 23, 2024		
Paid by Tyson via Watcard				
Description	Quantity	Unit Price	Measure	Total
Alum Plate 3/8 - 6061 T6	2.100	\$33.68	square foot	\$70.73
Alum Sheet - 5052 H32 1/8 - .125	2.600	\$12.69	square foot	\$32.99
	14.000	\$0.75	min	\$10.50
Labour	1.500	\$15.00	hour	\$22.50
			Sub Total	\$136.72
			Mat'l Tax (13%)	\$13.48
			Total	\$150.21
Order 5		Friday Mar 1, 2024		
Paid by Tyson via Watcard				
Description	Quantity	Unit Price	Measure	Total
M2 x 0.4 SHCS 10mm lg.	6.000	\$0.07	each	\$0.42
M3 x 0.5 SHCS 16mm lg.	6.000	\$0.09	each	\$0.54
M3 x 0.5 SHCS 25mm lg.	4.000	\$0.08	each	\$0.32
M3 x 0.5 SHCS 45mm lg.	4.000	\$0.18	each	\$0.72
M3 x 0.5 Set Screw 5 mm lg.	3.000	\$0.04	each	\$0.12
M3 Dowel Pin 36mm lg.	2.000	\$0.46	each	\$0.92
M4 x 0.7 FHCS 20mm lg.	4.000	\$0.19	each	\$0.76
M4 x 0.7 SHCS 10mm lg.	14.000	\$0.04	each	\$0.56
M4 x 0.7 SHCS 16mm lg.	12.000	\$0.08	each	\$0.96
M4 x 0.7 SHCS 20mm lg.	1.000	\$0.11	each	\$0.11
M6 x 1.0 SHCS 30mm lg.	2.000	\$0.32	each	\$0.64
			Sub Total	\$6.07
			Tax (13%)	\$0.79
			Total	\$6.86
Order 6		Monday Mar 4, 2024		
Paid by Tyson via Watcard				
Description	Quantity	Unit Price	Measure	Total
Sandpaper - 1000 grit	2.000	\$2.86	sheet	\$5.72
Sandpaper - 500 grit	2.000	\$0.92	sheet	\$1.84
Alum Sheet - 5052 H32 10 GA - .102	2.000	\$9.61	square foot	\$19.22
			Sub Total	\$26.78
			Tax (13%)	\$3.48
			Total	\$30.26

Order 7	Wednesday Mar 6, 2024			
Paid by Tyson via Watcard				
Description	Quantity	Unit Price	Measure	Total
Hex Head Plug 1/4 NPT	10.000	\$0.52	each	\$5.20
M3 x 0.5 SHCS 12mm lg.	40.000	\$0.03	each	\$1.20
M3 x 0.5 Hex Nut	20.000	\$0.02	each	\$0.40
Alum Flat 1/4 x 6 - 6061	1.000	\$10.92	foot	\$10.92
			Sub Total	\$17.72
			Tax (13%)	\$2.30
			Total	\$20.02
Order 8	Thursday Mar 7, 2024			
Paid by Tyson via Watcard				
Description	Quantity	Unit Price	Measure	Total
Alum Flat 3/8 x 6 - 6061	1.000	\$10.25	foot	\$10.25
			Sub Total	\$10.25
			Tax (13%)	\$1.33
			Total	\$11.58
Order 9	Monday Mar 11, 2024			
Paid by Tyson via Watcard				
Description	Quantity	Unit Price	Measure	Total
M3 x 0.5 FHCS 16mm lg.	10.000	\$0.25	each	\$2.50
M3 x 0.5 SHCS 12mm lg.	40.000	\$0.03	each	\$1.20
M3 x 0.5 SHCS 25mm lg.	15.000	\$0.08	each	\$1.20
M3 x 0.5 Hex Nut	10.000	\$0.02	each	\$0.20
M4 x 0.7 Threaded Rod	1.500	\$0.59	foot	\$0.89
Acrylic Sheet 1/16 - .0625	1.000	\$3.19	square foot	\$3.19
			Sub Total	\$9.18
			Tax (13%)	\$1.19
			Total	\$10.37
Grand Total (Material)	\$380.92			

10.1.4.2 3D Printing

Description	Date	Material	Payment Method	Who Paid	Unit Cost	Quantity	Total
Auger housing and Auger Prototype	07/11/2023	PLA	Wat Card	Justin	\$20.19	1	\$20.19
Bag Holder Assy (Backplate, clamp and 2 bushings)	02/07/2024	PLA	Wat Card	Kwaku	\$3.00	1	\$3.00
Auger housing and Auger Version 1	02/21/2024	PLA-CF	Cash	Tyson	\$15.00	1	\$15.00
Electrical Tower and Bag Feeder Lid	02/21/2024	PLA	Wat Card	Justin	\$18.67	1	\$18.67
Auger housing and Auger Version 2	02/21/2024	PLA	Wat Card	Kwaku	\$9.12	1	\$9.12
Bag Clamps	02/21/2024	PLA-CF	Wat Card	Tyson	\$3.00	4	\$12.00
						Total	\$38.19
						Tax incl	

10.1.4.3 Controls Components

Description	Date	Purchaser	Quantity	Unit Price	Measure	Sub Total	Tax	Total
10 Pcs SS Torsion Spring - 0.16" OD 90 deg	1/16/2024	Kwaku	1.000	\$19.98	Each	\$19.98	\$2.60	\$22.58
40mm ID PVC Ball Valve	1/16/2024	Kwaku	1.000	\$22.99	Each	\$22.99	\$2.99	\$25.98
6908-2RS Deep Groove Ball Bearing	1/16/2024	Kwaku	1.000	\$10.06	Each	\$10.06	\$1.31	\$11.37
1.5" ID Gate Valve	1/16/2024	Kwaku	1.000	\$24.91	Each	\$24.91	\$3.24	\$28.15
10 Pcs SS Torsion Spring - 50 mm 90 deg	1/22/2024	Justin	1.000	\$17.65	Each	\$17.65	\$2.29	\$19.94
Aluminum 25T Servo Horn (2-Pack)	1/31/2024	Justin	1.000	\$12.97	Each	\$12.97	\$1.69	\$14.66
30 kg High Speed Servo Motor	1/31/2024	Justin	1.000	\$25.97	Each	\$25.97	\$3.38	\$29.35
Aluminum Heating Plate	2/1/2024	Kwaku	1.000	\$17.33	Each	\$17.33	\$2.25	\$19.58
608-2RS Deep Groove Ball Bearing 10 pcs	2/5/2024	Justin	1.000	\$12.59	Each	\$12.59	\$1.64	\$14.23
Neodymium Magnets	2/13/2024	Justin	1.000	\$12.77	Each	\$12.77	\$1.66	\$14.43
304 Stainless Rod	2/22/2024	Justin	1.000	\$15.99	Each	\$15.99	\$2.08	\$18.07
8mm Bushing	2/22/2024	Justin	1.000	\$13.14	Each	\$13.14	\$1.71	\$14.85
Mosfet Board	2/22/2024	Justin	2.000	\$11.19	Each	\$22.38	\$2.91	\$25.29
Vacuum Generator	2/22/2024	Justin	2.000	\$13.07	Each	\$26.14	\$3.40	\$29.54
12V Solenoid 2/5Way Pneumatic Valve Bank	2/22/2024	Justin	1.000	\$78.59	Each	\$78.59	\$10.22	\$88.81
100mm Air Cylinder	2/22/2024	Justin	1.000	\$15.71	Each	\$15.71	\$2.04	\$17.75
Elegoo Mega 2560 R3 Board	2/23/2024	Justin	1.000	\$29.95	Each	\$29.95	\$3.89	\$33.84
12V 29Amp AC/DC Power Supply	2/23/2024	Kwaku	1.000	\$50.00	Each	\$50.00	\$6.50	\$56.50
200mm Linear Rail	2/23/2024	Kwaku	1.000	\$20.99	Each	\$20.99	\$2.73	\$23.72
12V to 36V Step Up Transformer	2/23/2024	Kwaku	1.000	\$31.95	Each	\$31.95	\$4.15	\$36.10
Power Distribution Module	2/23/2024	Kwaku	1.000	\$61.28	Each	\$61.28	\$7.97	\$69.25
12V to 5V Step Down Voltage Regulator	2/23/2024	Kwaku	1.000	\$19.99	Each	\$19.99	\$2.60	\$22.59
Rubber Bumper Feet	3/4/2024	Tyson	1.000	\$9.99	Each	\$9.99	\$1.30	\$11.29
Assorted Spring Kit	3/4/2024	Tyson	1.000	\$15.99	Each	\$15.99	\$2.08	\$18.07
100mm Linear Rail	3/4/2024	Tyson	1.000	\$8.44	Each	\$8.44	\$1.10	\$9.54
Cable Zip Ties	3/8/2024	Kwaku	1.000	\$14.99	Each	\$14.99	\$1.95	\$16.94
Braided Cable Sleeve	3/8/2024	Kwaku	1.000	\$14.99	Each	\$14.99	\$1.95	\$16.94
20 AWG Solid Core Wire	3/8/2024	Kwaku	1.000	\$21.99	Each	\$21.99	\$2.86	\$24.85
100mm Linear Rail (#2)	3/10/2024	Tyson	1.000	\$20.69	Each	\$20.69	\$2.69	\$23.38
Nema 17 42-48 Stepper Motor	3/10/2024	Tyson	2.000	\$24.99	Each	\$49.98	\$6.50	\$56.48
150mm Linear Rail	3/10/2024	Tyson	1.000	\$20.33	Each	\$20.33	\$2.64	\$22.97
TB6600 Stepper Motor Driver	3/10/2024	Tyson	1.000	\$27.99	Each	\$27.99	\$3.64	\$31.63
50mm Air Cylinder	3/11/2024	Justin	1.000	\$15.07	Each	\$15.07	\$1.96	\$17.03
Braided Cable Sleeve	3/14/2024	Kwaku	1.000	\$14.99	Each	\$14.99	\$1.95	\$16.94
Power Switch	3/18/2024	Kwaku	1.000	\$14.28	Each	\$14.28	\$1.86	\$16.14
Thermocouple Sensor	3/23/2024	Kwaku	1.000	\$17.99	Each	\$17.99	\$2.34	\$20.33
							Total	\$939.08

10.1.5 Risk Register and Mitigation Plan

Table 2: Risk identification and mitigation table

Risk	Consequences	Mitigation
Fail to design a subsystem that can accurately measure quantities of various seeds	Fail to meet the most fundamental design constraint of the project. Results in complete failure of project.	Make this subsystem the first to enter the design loop. This will allow most of time to be dedicated to it, and minimize the design constraints based on preexisting subsystems.
Seeds get stuck to surfaces within the device restricting them from moving accurately.	Limit the accuracy/ efficiency of measuring quantities of seeds. Lead to waste of seeds as well as cross contamination.	Test early on different materials against all types of seeds to evaluate if any combinations have problems with sticking.
The seed packet's flimsy structure limits consistent handling throughout the device.	Less efficient device, waste of packets and possibly seeds, and possibility of jammed system that requires maintenance.	Design clamp system to have multiple points of contact with packets.
Varying sides of seeds varies the performance of different subsystems.(ex: different flow rate out of hopper)	Possibility of inaccurate measuring.	When completing the controls of the system, test every seed independently. Create independent programs for each seed and have the user enter the seed type before beginning the process.
Not having enough time to properly design each subsystem and integrate.	Limits the scope of our project, possibly failing to deliver on core restraints.	Start with the most important subsystems that deliver our core design constraints. Then expand the scope. Build in extra time to schedule.
Design does not work with obscure seed types. Not spherical, very large, and very small.	Limits the capabilities of the device.	Begin designing the device to handle the majority of seed types (spherical and size of apple seeds) then with time permitting adapt the system to handle obscure seeds.

10.2 Engineering Data

10.2.1 Electrical System Power

Table 3: Electrical system total power

System			Electrical Requirements		
Subsystem	Component	Interface	Voltage [V]	Current [A]	Power [W]
Heat Sealer	Heat Element	Step-up Transformer	12	6	72
	1-way Valve	MOSFET	12	0.54	6.48
Packet Ejection	1-way Valve	MOSFET	12	0.54	6.48
	2-way Valve	MOSFET	12	0.54	6.48
Packet Injection	1-way Valve	MOSFET	12	0.54	6.48
	2-way Valve	MOSFET	12	0.54	6.48
Controls	Arduino		12	0.085	1.02
Rotary Base	Stepper	DRV8825	12	1.9	22.8
Seed Portioner	Stepper	DRV8825	12	1.9	22.8
	Servo	Step-down Transformer	12	0.75	9
	2-way Valve	MOSFET	12	0.54	6.48
	1-way Valve	MOSFET	12	0.54	6.48
Total				14.415	172.98

10.3 Engineering Specification

Table 4: Engineering specification table

No.	Characteristic	Relation	Value	Units	Verification Method	Comments
1	Fill accuracy	+/-	10	% weight	Scale	
2	Production rate	>=	15	Packets/min	Timer	
3	Set-up time	>=	120	Seconds	Timer	
4	Production amount	<=	100	Packets	Count	
5	Failure rate	<=	1/1000	Packets	Count	Conduct trial runs to determine failure rate.
6	Self-cleaning	Pass/fail	--	--	Observation	Check for dust/debris particles.
7	Hopper storage	=	1.5	L	Water measurement	
8	Seed temperature	<	25	Celsius	Infrared video	
9	Maintenance frequency	<	1	Hour/month	Observation	Conduct trial runs to determine maintenance requirements.
10	Corrosion/wear resistance	Pass/fail	--	--	Observation	Conduct trial runs to determine equipment wear.

10.4 Verification of Design Data

10.4.1 Seed portioning verification

Table 5: Seed portioning verification results

Repetition	Number of skittles
1	5
2	7
3	7
4	8
5	6
6	7
7	8
8	8
9	7
10	7
11	8
12	7
13	7
14	7
15	8
16	9
17	8
18	7
19	7
20	7
21	6
22	8
23	9
24	7
25	8
26	7
27	8
28	7
29	8
30	7
31	7
32	7
33	6
34	7
35	7
36	7
37	7
38	7
39	8
40	8

10.5 Hazard Disclosure Form

Team Number:

22

Team Project:

Seed Packaging System

Faculty Advisor(s):

Mohammed Nassar

Brief Project Description (2-3 sentences):

The purpose of this system is to automate the process of separating and packaging bulk seeds into smaller denominations. This process is currently being done by hand which results in long processing times with high effort. This system will be controlled by an arduino, utilizing stepper motors, pneumatics and custom processing mechanisms to fully automate the packaging process.

Table 6: Hazard Identification and Mitigation:

Hazard	Identification and mitigation
Electricity -	Electricity will be used to power pneumatic solenoids and packet sealer (heater). Proper wiring will be done with insulation and guarding over any terminals.
Heat -	The packet sealer will use a heater element to melt the packet together. Shielding will be used to prevent burns from accidental contact.
Compressed Gas -	Compressed air will be used to actuate components. Only off the shelf pneumatic components will be utilized with appropriate safety ratings and relief valves.
Biological Components -	The system processes seeds which are biological objects. We don't see any risks associated with this.
Ejected Gases -	Compressed air will be exhausted from pneumatic actuators. Mufflers will be used to reduce noise levels and reduce flow pressure and strength.
Moving Parts -	Numerous moving parts exist within the system. Shielding will be used to prevent access to potential pinch points.